

Parallel and Perpendicular Lines

Worked Solutions with TikZ Graphs

Wisest Maths — Year 1 A-Level, Coordinate Geometry (ref `cg3`)

This document contains all 70 *Parallel and Perpendicular Lines* questions, each with a fully worked solution and (where a single plot is meaningful) a TikZ diagram on the coordinate plane: the given line is drawn in grey and the answer line in blue, with parallel lines sharing a gradient and perpendicular lines meeting at a right angle (the axes use equal scaling so right angles look right). Perpendicular bisectors, altitudes, feet of perpendiculars, triangles and quadrilaterals are shown too.

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Parallel and Perpendicular Lines 01 (cg3-001)

Difficulty: Foundation / Marks: 3

State which of the following lines are parallel to $y = 4x - 3$.

a) $2y = 8x + 5$ b) $y - 4x + 1 = 0$ c) $8y + 32x = 1$ d) $\frac{1}{2}(y + 2) = 2x$ e) $-12y - 3 = -48x$ f) $4y = x$

Worked solution.

Step 1: A line is parallel to $y = 4x - 3$ if and only if its gradient is 4. Rearrange each into $y = mx + c$ form.

$$\text{a) } 2y = 8x + 5 \implies y = 4x + 2.5 \quad m = 4 \checkmark$$

Divide every term by 2 to isolate y . The coefficient of x is the gradient.

$$\text{b) } y = 4x - 1 \quad m = 4 \checkmark$$

Move $-4x + 1$ to the right-hand side: $y = 4x - 1$.

$$\text{c) } 8y = -32x + 1 \implies y = -4x + \frac{1}{8} \quad m = -4 \times$$

Subtract $32x$ from both sides, then divide by 8. Gradient is -4 , not 4, so not parallel.

$$\text{d) } \frac{1}{2}y + 1 = 2x \implies y = 4x - 2 \quad m = 4 \checkmark$$

Distribute the $\frac{1}{2}$ on the left to get $\frac{1}{2}y + 1 = 2x$, then multiply through by 2 to clear the fraction.

$$\text{e) } -12y = -48x + 3 \implies y = 4x - \frac{1}{4} \quad m = 4 \checkmark$$

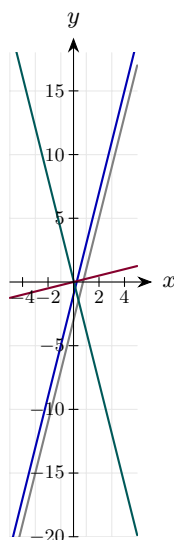
Divide every term by -12 . Be careful with signs: $\frac{-48}{-12} = 4$ and $\frac{3}{-12} = -\frac{1}{4}$.

$$\text{f) } y = \frac{1}{4}x \quad m = \frac{1}{4} \times$$

Divide both sides by 4. Gradient is $\frac{1}{4}$, not 4 — a common slip is to mix up $4y = x$ with $y = 4x$.

Final answer: Lines a), b), d) and e) are parallel to $y = 4x - 3$.

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 02 (cg3-002)

Difficulty: Foundation / Marks: 4

State whether each pair of lines is parallel, perpendicular, or neither.

a) $y = 3x + 1$ and $y = 3x - 5$ b) $y = 2x + 4$ and $y = -\frac{1}{2}x + 1$ c) $y = 5x - 2$ and $y = -5x + 2$ d) $2x - 4y + 3 = 0$ and $x - 2y - 1 = 0$

Worked solution.

Step 1: Compare gradients. Two lines are parallel if $m_1 = m_2$; perpendicular if $m_1 \times m_2 = -1$.

$$\text{a) } m_1 = 3, m_2 = 3 \implies \text{parallel}$$

Both lines are already in $y = mx + c$ form, so the gradients can be read straight off. Equal gradients (and different intercepts) means parallel.

$$\text{b) } m_1 = 2, m_2 = -\frac{1}{2}, 2 \times \left(-\frac{1}{2}\right) = -1 \implies \text{perpendicular}$$

$-\frac{1}{2}$ is the negative reciprocal of 2, so the product of gradients is -1 — the perpendicularity condition.

$$\text{c) } m_1 = 5, m_2 = -5, 5 \times (-5) = -25 \neq -1 \implies \text{neither}$$

A common mistake is to assume that opposite signs alone imply perpendicularity. For perpendicular lines you need the negative *reciprocal* (here $-\frac{1}{5}$), not just the negative.

Step 4: Rearrange part d): both give $m = \frac{1}{2}$.

$$2x - 4y + 3 = 0 \implies y = \frac{1}{2}x + \frac{3}{4}; \quad x - 2y - 1 = 0 \implies y = \frac{1}{2}x - \frac{1}{2} \implies \text{parallel}$$

Isolate y in each equation by moving the x and constant terms across, then divide by the coefficient of y . Equal gradients with different intercepts confirm parallel.

Final answer: a) Parallel; b) Perpendicular; c) Neither; d) Parallel

Parallel and Perpendicular Lines 03 (cg3-003)

Difficulty: Foundation / Marks: 3

State whether each pair of lines is parallel, perpendicular, or neither.

a) $3x + y - 4 = 0$ and $6x + 2y + 1 = 0$ **b)** $4x - y + 2 = 0$ and $x + 4y - 8 = 0$ **c)** $2x + 5y - 10 = 0$ and $5x - 2y + 3 = 0$

Worked solution.

Step 1: Rearrange to find gradients.

$$\text{a) } y = -3x + 4 \text{ and } y = -3x - \frac{1}{2} \implies m_1 = m_2 = -3 \implies \text{parallel}$$

For the general form $Ax + By + C = 0$ the gradient is $-\frac{A}{B}$. Both lines give $m = -3$, so they are parallel.

$$\text{b) } y = 4x + 2 \text{ and } y = -\frac{1}{4}x + 2 \implies 4 \times \left(-\frac{1}{4}\right) = -1 \implies \text{perpendicular}$$

$-\frac{1}{4}$ is the negative reciprocal of 4, so the product is -1 . Note both lines pass through $(0, 2)$, so they meet at a right angle there.

$$\text{c) } y = -\frac{2}{5}x + 2 \text{ and } y = \frac{5}{2}x + \frac{3}{2} \implies \left(-\frac{2}{5}\right) \times \frac{5}{2} = -1 \implies \text{perpendicular}$$

The gradients $-\frac{2}{5}$ and $\frac{5}{2}$ are negative reciprocals (flip the fraction and swap sign), so their product is -1 .

Final answer: a) Parallel; b) Perpendicular; c) Perpendicular

Parallel and Perpendicular Lines 04 (cg3-004)

Difficulty: Foundation / Marks: 3

Find the value of k so that the line $kx + 3y - 5 = 0$ is parallel to $y = 4x - 2$.

Worked solution.

Step 1: Rearrange $kx + 3y - 5 = 0$ to find its gradient.

$$3y = -kx + 5 \implies y = -\frac{k}{3}x + \frac{5}{3}$$

Move the x and constant terms to the right, then divide through by the coefficient of y . The coefficient of x is the gradient, here $-\frac{k}{3}$.

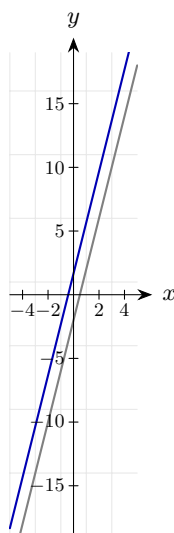
Step 2: For parallel lines, set gradients equal: $-\frac{k}{3} = 4$.

$$-\frac{k}{3} = 4 \implies k = -12$$

Multiply both sides by -3 . Watch the sign — multiplying by a negative flips the right-hand side from 4 to -12 .

Final answer: $k = -12$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 05 (cg3-005)

Difficulty: Foundation / Marks: 3

Find the equation of the line parallel to $y = 3x - 1$ that passes through $(2, 8)$. Give your answer in the form $y = mx + c$.

Worked solution.

Step 1: The gradient of the parallel line is $m = 3$. Write $y = 3x + c$.

$$y = 3x + c$$

Parallel lines share the same gradient.

Step 2: Substitute $(2, 8)$ to find c .

$$8 = 3(2) + c \implies c = 2$$

The line must pass through $(2, 8)$, so these coordinates satisfy the equation. Solve for c to pin down the y -intercept.

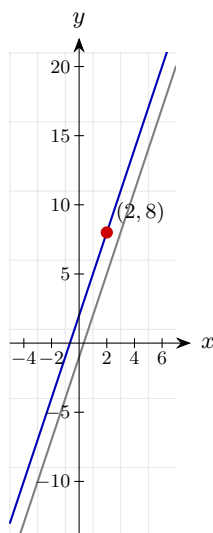
Step 3: Write the equation.

$$y = 3x + 2$$

Substitute the gradient and c back into $y = mx + c$.

Final answer: $y = 3x + 2$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 06 (cg3-006)

Difficulty: Foundation / Marks: 4

Find the equation of the line parallel to $y = -2x + 5$ passing through $(3, 1)$. Give your answer in the form $ax + by + c = 0$.

Worked solution.

Step 1: The gradient is $m = -2$. Use $y = -2x + c$ and substitute $(3, 1)$.

$$1 = -2(3) + c \implies c = 7$$

Parallel lines share gradients, so $m = -2$. Plug the given point in to solve for c : $1 = -6 + c$ gives $c = 7$.

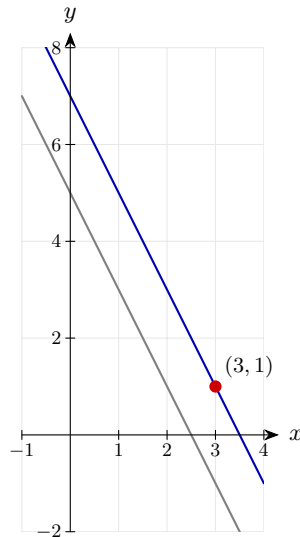
Step 2: Rearrange $y = -2x + 7$ into $ax + by + c = 0$.

$$2x + y - 7 = 0$$

Move every term to one side. Conventionally we take the coefficient of x positive, so add $2x$ and subtract 7 from both sides.

Final answer: $2x + y - 7 = 0$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 07 (cg3-007)

Difficulty: Foundation / Marks: 4

Line L passes through $(0, 5)$ and is parallel to $3x - 6y + 4 = 0$. Find the equation of L in the form $y = mx + c$.

Worked solution.

Step 1: Rearrange $3x - 6y + 4 = 0$ to find its gradient.

$$6y = 3x + 4 \implies y = \frac{1}{2}x + \frac{2}{3}$$

Gradient is $m = \frac{1}{2}$.

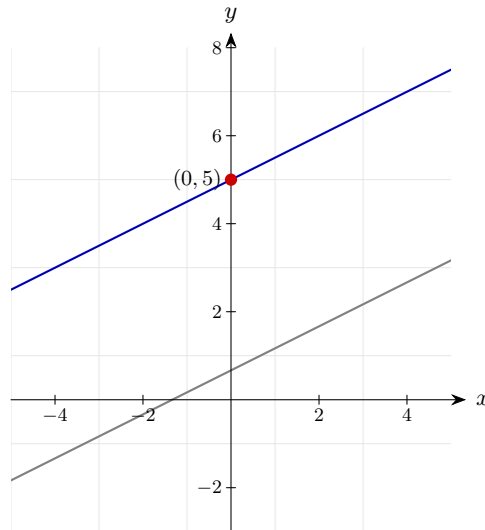
Step 2: Since L passes through $(0, 5)$, the y -intercept is 5.

$$y = \frac{1}{2}x + 5$$

When $x = 0$, $y = c$, so $c = 5$ directly.

Final answer: $y = \frac{1}{2}x + 5$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 08 (cg3-008)

Difficulty: Foundation / Marks: 4

Find the equation of the line parallel to $5x + 2y - 6 = 0$ that passes through $(4, -3)$. Give your answer in the form $ax + by + c = 0$, where a , b and c are integers.

Worked solution.

Step 1: Find the gradient of the given line.

$$2y = -5x + 6 \implies y = -\frac{5}{2}x + 3 \implies m = -\frac{5}{2}$$

Move the $5x$ and -6 across, then divide by 2. The parallel line will share this gradient.

Step 2: Use $y = -\frac{5}{2}x + c$ with point $(4, -3)$.

$$-3 = -\frac{5}{2}(4) + c \implies -3 = -10 + c \implies c = 7$$

Substitute the point into the parallel line equation. $-\frac{5}{2} \times 4 = -10$, then add 10 to both sides.

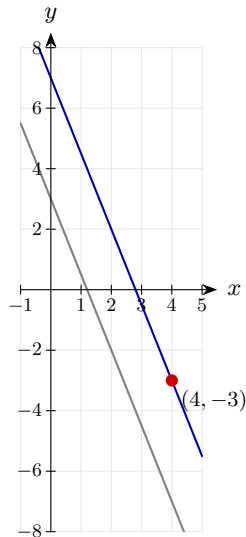
Step 3: Rearrange $y = -\frac{5}{2}x + 7$: multiply by 2.

$$2y = -5x + 14 \implies 5x + 2y - 14 = 0$$

Multiplying by 2 clears the fraction so the coefficients are integers, as required by the question.

Final answer: $5x + 2y - 14 = 0$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 09 (cg3-009)

Difficulty: Foundation / Marks: 4

Find the equation of the line parallel to $4x - 3y + 9 = 0$ passing through $(-3, 2)$. Give your answer in the form $ax + by + c = 0$.

Worked solution.

Step 1: Find gradient: $3y = 4x + 9 \implies m = \frac{4}{3}$.

$$m = \frac{4}{3}$$

Rearrange the given line into $y = mx + c$ form by isolating y . The parallel line will have the same gradient.

Step 2: Use $y = \frac{4}{3}x + c$ with $(-3, 2)$.

$$2 = \frac{4}{3}(-3) + c \implies 2 = -4 + c \implies c = 6$$

Substituting $x = -3$ into $\frac{4}{3}x$ gives -4 (the 3s cancel). Then $c = 2 + 4 = 6$.

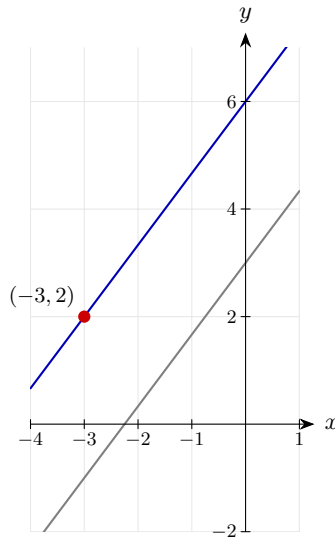
Step 3: Rearrange $y = \frac{4}{3}x + 6$: multiply by 3.

$$3y = 4x + 18 \implies 4x - 3y + 18 = 0$$

Multiplying by 3 clears the fraction. Then bring everything to one side, keeping the x -coefficient positive.

Final answer: $4x - 3y + 18 = 0$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 10 (cg3-010)

Difficulty: Foundation / Marks: 4

Line P passes through $(6, -1)$ and is parallel to the line joining $(0, 3)$ and $(4, 11)$. Find the equation of P in the form $y = mx + c$.

Worked solution.

Step 1: Find the gradient of the line joining $(0, 3)$ and $(4, 11)$.

$$m = \frac{11 - 3}{4 - 0} = \frac{8}{4} = 2$$

Use the gradient formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. P is parallel to this line, so it shares the gradient.

Step 2: Use $y = 2x + c$ with $(6, -1)$.

$$-1 = 2(6) + c \implies c = -13$$

Substitute the point on P . $c = -1 - 12 = -13$.

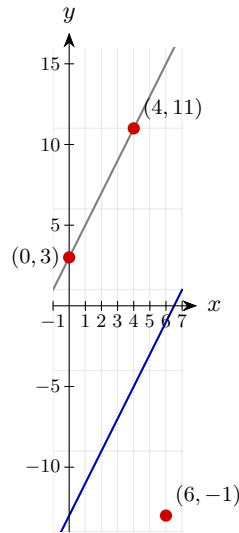
Step 3: Write the equation.

$$y = 2x - 13$$

Combine the gradient and y -intercept to give the final equation.

Final answer: $y = 2x - 13$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 11 (cg3-011)

Difficulty: Foundation / Marks: 5

Find the equation of the line parallel to $\frac{x}{3} + \frac{y}{2} = 1$ that passes through $(6, 5)$. Give your answer in the form $ax + by + c = 0$.

Worked solution.

Step 1: Rearrange the given line to find its gradient.

$$\frac{x}{3} + \frac{y}{2} = 1 \implies 2x + 3y = 6 \implies y = -\frac{2}{3}x + 2$$

Multiply through by 6 to clear fractions, then isolate y .

Step 2: The parallel line has gradient $m = -\frac{2}{3}$. Substitute $(6, 5)$.

$$5 = -\frac{2}{3}(6) + c \implies 5 = -4 + c \implies c = 9$$

$-\frac{2}{3} \times 6 = -4$ (the 3s cancel). Then add 4 to both sides to find c .

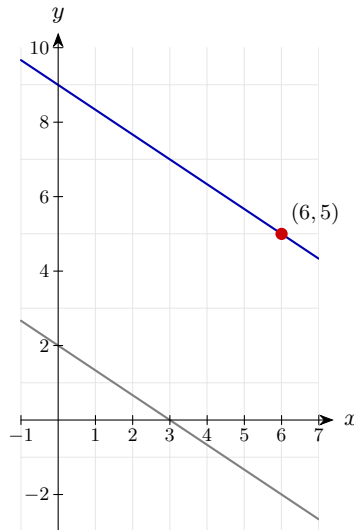
Step 3: Rearrange $y = -\frac{2}{3}x + 9$: multiply by 3.

$$3y = -2x + 27 \implies 2x + 3y - 27 = 0$$

Multiply by 3 to clear the fraction, then move all terms to one side to get the required integer-coefficient form.

Final answer: $2x + 3y - 27 = 0$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 12 (cg3-012)

Difficulty: Foundation / Marks: 5

Two parallel lines each have gradient $\frac{3}{2}$. One passes through $(2, 4)$ and the other through $(6, 3)$. Find the equations of both lines in the form $ax + by + c = 0$.

Worked solution.

Step 1: Line 1 through $(2, 4)$ with $m = \frac{3}{2}$:

$$4 = \frac{3}{2}(2) + c \implies c = 1 \implies y = \frac{3}{2}x + 1 \implies 3x - 2y + 2 = 0$$

Substitute the point to find c , then multiply through by 2 and rearrange to $ax + by + c = 0$ form. Multiplying $y = \frac{3}{2}x + 1$ by 2 gives $2y = 3x + 2$.

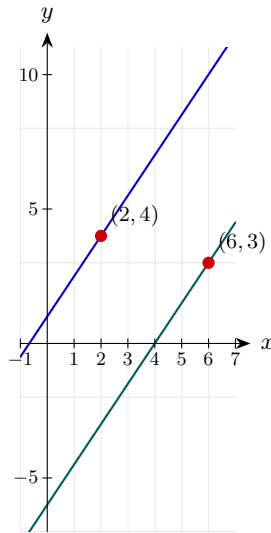
Step 2: Line 2 through $(6, 3)$ with $m = \frac{3}{2}$:

$$3 = \frac{3}{2}(6) + c \implies c = -6 \implies y = \frac{3}{2}x - 6 \implies 3x - 2y - 12 = 0$$

Same method: $\frac{3}{2}(6) = 9$, so $c = 3 - 9 = -6$. Multiply by 2 and rearrange to integer form.

Final answer: Line 1: $3x - 2y + 2 = 0$; Line 2: $3x - 2y - 12 = 0$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 13 (cg3-013)

Difficulty: Foundation / Marks: 3

Find the equation of the line perpendicular to $y = 2x - 1$ that passes through $(4, 3)$. Give your answer in the form $y = mx + c$.

Worked solution.

Step 1: The gradient of $y = 2x - 1$ is 2. The perpendicular gradient is $-\frac{1}{2}$.

$$m_{\perp} = -\frac{1}{2}$$

Perpendicular gradients multiply to -1 .

Step 2: Substitute $(4, 3)$ into $y = -\frac{1}{2}x + c$.

$$3 = -\frac{1}{2}(4) + c \implies c = 5$$

$-\frac{1}{2} \times 4 = -2$, so $c = 3 + 2 = 5$.

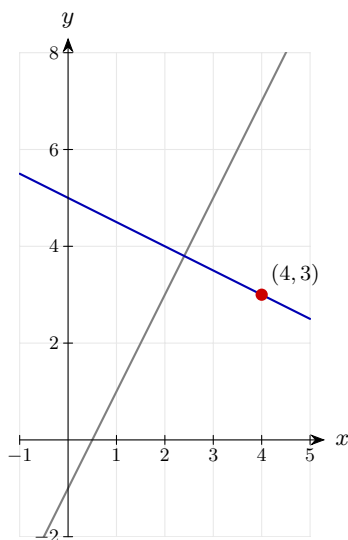
Step 3: Write the equation.

$$y = -\frac{1}{2}x + 5$$

Put the gradient and y -intercept together.

Final answer: $y = -\frac{1}{2}x + 5$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 14 (cg3-014)

Difficulty: Foundation / Marks: 3

**Find the equation of the line perpendicular to $y = -\frac{1}{3}x + 4$ that passes through $(1, -2)$.
Give your answer in the form $y = mx + c$.**

Worked solution.

Step 1: The given gradient is $-\frac{1}{3}$. Perpendicular gradient is 3.

$$m_{\perp} = -1 \div \left(-\frac{1}{3}\right) = 3$$

The perpendicular gradient is the negative reciprocal. Dividing -1 by a negative gives a positive — flipping the fraction $-\frac{1}{3}$ and changing sign gives $+3$.

Step 2: Substitute $(1, -2)$ into $y = 3x + c$.

$$-2 = 3(1) + c \implies c = -5$$

Substitute the given point. $c = -2 - 3 = -5$.

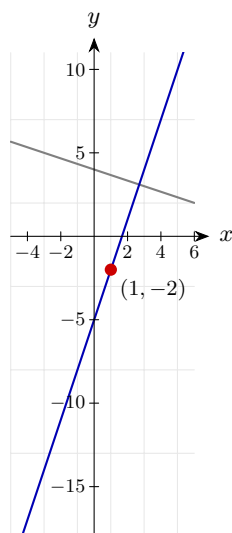
Step 3: Write the equation.

$$y = 3x - 5$$

Combine gradient and intercept.

Final answer: $y = 3x - 5$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 15 (cg3-015)

Difficulty: Foundation / Marks: 4

Find the equation of the line perpendicular to $4x - y + 3 = 0$ that passes through $(8, 1)$. Give your answer in the form $ax + by + c = 0$.

Worked solution.

Step 1: Rearrange $4x - y + 3 = 0$: $y = 4x + 3$, so $m = 4$.

$$m_{\perp} = -\frac{1}{4}$$

The negative reciprocal of 4 is $-\frac{1}{4}$, which is the gradient of any line perpendicular to the given one.

Step 2: Substitute $(8, 1)$ into $y = -\frac{1}{4}x + c$.

$$1 = -\frac{1}{4}(8) + c \implies 1 = -2 + c \implies c = 3$$

$-\frac{1}{4} \times 8 = -2$, so adding 2 to both sides gives $c = 3$.

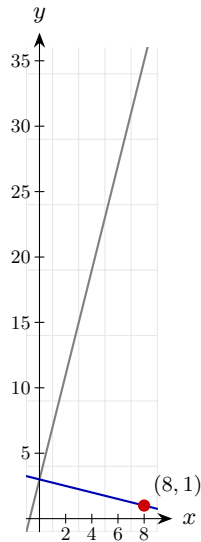
Step 3: Rearrange $y = -\frac{1}{4}x + 3$: multiply by 4.

$$4y = -x + 12 \implies x + 4y - 12 = 0$$

Multiplying by 4 clears the fraction. Bring everything to one side, keeping the x -coefficient positive.

Final answer: $x + 4y - 12 = 0$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 16 (cg3-016)

Difficulty: Foundation / Marks: 5

Find the equation of the line perpendicular to $3x + 5y - 2 = 0$ that passes through $(-5, 4)$. Give your answer in the form $ax + by + c = 0$.

Worked solution.

Step 1: Rearrange: $5y = -3x + 2 \implies y = -\frac{3}{5}x + \frac{2}{5}$, so $m = -\frac{3}{5}$.

$$m_{\perp} = -1 \div \left(-\frac{3}{5}\right) = \frac{5}{3}$$

Flip the fraction $-\frac{3}{5}$ and change sign to get the perpendicular gradient $\frac{5}{3}$.

Step 2: Substitute $(-5, 4)$ into $y = \frac{5}{3}x + c$.

$$4 = \frac{5}{3}(-5) + c \implies 4 = -\frac{25}{3} + c \implies c = \frac{37}{3}$$

Convert 4 to thirds: $4 = \frac{12}{3}$, so $c = \frac{12}{3} + \frac{25}{3} = \frac{37}{3}$.

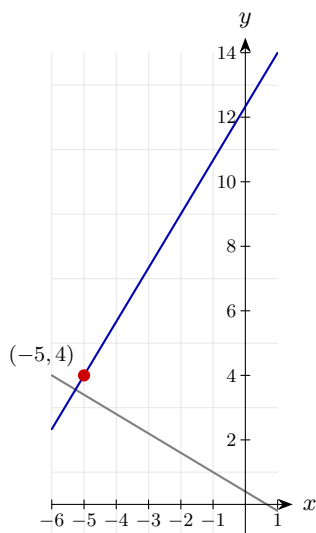
Step 3: Rearrange $y = \frac{5}{3}x + \frac{37}{3}$: multiply by 3.

$$3y = 5x + 37 \implies 5x - 3y + 37 = 0$$

Multiplying by 3 clears both fractions in one step. Then rearrange to the required integer form.

Final answer: $5x - 3y + 37 = 0$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 17 (cg3-017)

Difficulty: Foundation / Marks: 4

Find the equation of the line perpendicular to $2x + 6y - 3 = 0$ passing through $(3, -4)$. Give your answer in the form $y = mx + c$.

Worked solution.

Step 1: Rearrange: $6y = -2x + 3 \implies m = -\frac{1}{3}$.

$$m_{\perp} = 3$$

Dividing through by 6 gives gradient $-\frac{2}{6} = -\frac{1}{3}$. The negative reciprocal is 3.

Step 2: Substitute $(3, -4)$ into $y = 3x + c$.

$$-4 = 3(3) + c \implies c = -13$$

Plug in the point: $c = -4 - 9 = -13$.

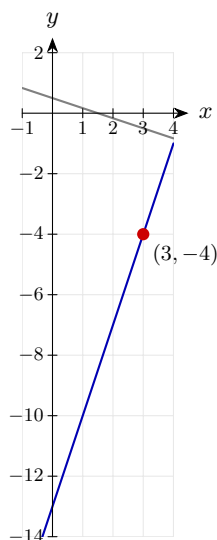
Step 3: Write the equation.

$$y = 3x - 13$$

Combine the gradient and y -intercept.

Final answer: $y = 3x - 13$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 18 (cg3-018)

Difficulty: Foundation / Marks: 5

Line M is perpendicular to the line joining $A(1, 5)$ and $B(4, 11)$, and passes through the midpoint of AB . Find the equation of M in the form $y = mx + c$.

Worked solution.

Step 1: Find the gradient of AB .

$$m_{AB} = \frac{11 - 5}{4 - 1} = \frac{6}{3} = 2$$

Apply the gradient formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ to the two given points.

Step 2: The perpendicular gradient is $m_{\perp} = -\frac{1}{2}$.

$$m_{\perp} = -\frac{1}{2}$$

Negative reciprocal of 2.

Step 3: Find the midpoint of AB .

$$M = \left(\frac{1 + 4}{2}, \frac{5 + 11}{2} \right) = (2.5, 8)$$

The midpoint is the average of the endpoint coordinates. M must pass through this point.

Step 4: Substitute $(2.5, 8)$ into $y = -\frac{1}{2}x + c$.

$$8 = -\frac{1}{2}(2.5) + c \implies c = 9.25$$

$-\frac{1}{2} \times 2.5 = -1.25$, so $c = 8 + 1.25 = 9.25$.

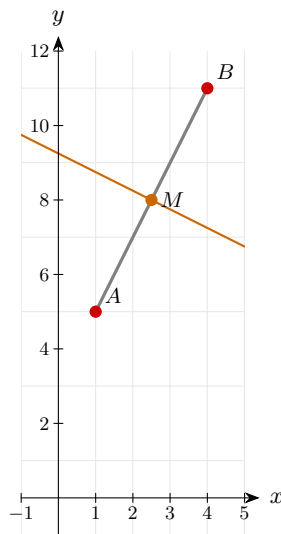
Step 5: Write the equation.

$$y = -\frac{1}{2}x + \frac{37}{4}$$

$9.25 = \frac{37}{4}$.

Final answer: $y = -\frac{1}{2}x + \frac{37}{4}$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 19 (cg3-019)

Difficulty: Foundation / Marks: 3

Find the value of k such that the line $kx - 2y + 5 = 0$ is perpendicular to $y = 4x - 1$.

Worked solution.

Step 1: Rearrange $kx - 2y + 5 = 0$: $y = \frac{k}{2}x + \frac{5}{2}$, gradient $= \frac{k}{2}$.

$$m_1 = \frac{k}{2}$$

Move $kx + 5$ across (sign flips), then divide by 2. Coefficient of x is the gradient.

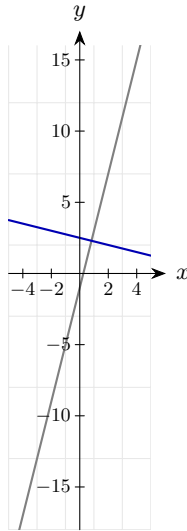
Step 2: For perpendicularity: $\frac{k}{2} \times 4 = -1$.

$$2k = -1 \implies k = -\frac{1}{2}$$

Multiplying the gradients gives $\frac{4k}{2} = 2k$. Setting this equal to -1 and dividing by 2 gives $k = -\frac{1}{2}$.

Final answer: $k = -\frac{1}{2}$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 20 (cg3-020)

Difficulty: Foundation / Marks: 5

Find the equation of the line that is perpendicular to $6x - 4y + 1 = 0$ and passes through the point $(3, 7)$. Give your answer in the form $ax + by + c = 0$.

Worked solution.

Step 1: Rearrange: $4y = 6x + 1 \implies y = \frac{3}{2}x + \frac{1}{4}$, so $m = \frac{3}{2}$.

$$m_{\perp} = -\frac{2}{3}$$

Negative reciprocal of $\frac{3}{2}$: flip to $\frac{2}{3}$, change sign.

Step 2: Substitute $(3, 7)$ into $y = -\frac{2}{3}x + c$.

$$7 = -\frac{2}{3}(3) + c \implies c = 9$$

$-\frac{2}{3} \times 3 = -2$, so $c = 7 + 2 = 9$.

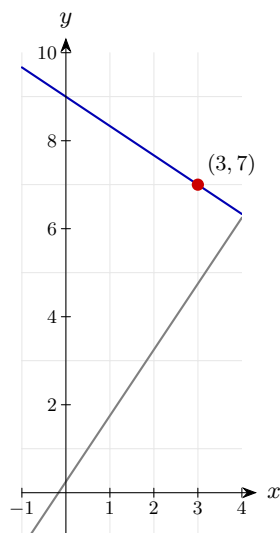
Step 3: Rearrange $y = -\frac{2}{3}x + 9$: multiply by 3.

$$3y = -2x + 27 \implies 2x + 3y - 27 = 0$$

Multiplying by 3 clears the fraction; then collect all terms on one side.

Final answer: $2x + 3y - 27 = 0$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 21 (cg3-021)

Difficulty: Foundation / Marks: 4

Line Q passes through (a, b) and is perpendicular to $5x - 2y = 10$. Find an equation for Q in terms of a and b .

Worked solution.

Step 1: Rearrange the given line: $y = \frac{5}{2}x - 5$, so $m = \frac{5}{2}$.

$$m_{\perp} = -\frac{2}{5}$$

Negative reciprocal of $\frac{5}{2}$ is $-\frac{2}{5}$.

Step 2: Use point (a, b) and gradient $-\frac{2}{5}$.

$$y - b = -\frac{2}{5}(x - a)$$

Point-slope form. This is the cleanest starting form when the point is given as algebraic letters rather than numbers.

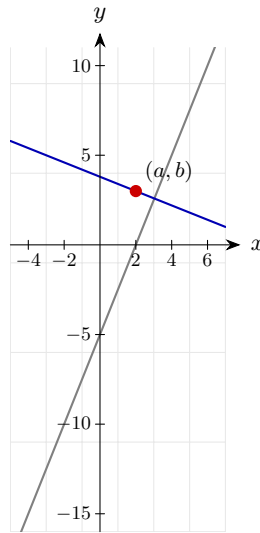
Step 3: Multiply through by 5 and rearrange.

$$5y - 5b = -2x + 2a \implies 2x + 5y = 2a + 5b$$

Multiplying by 5 clears the fraction. Then bring the x -term to the left to give a tidy form with integer coefficients of x and y .

Final answer: $2x + 5y = 2a + 5b$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 22 (cg3-022)

Difficulty: Foundation / Marks: 4

The line L_1 has equation $y = \frac{2}{3}x + 1$. Line L_2 is perpendicular to L_1 and intersects the x -axis at $x = 6$. Find the equation of L_2 .

Worked solution.

Step 1: The gradient of L_2 is the negative reciprocal of $\frac{2}{3}$.

$$m_{\perp} = -\frac{3}{2}$$

Flip the fraction and change sign.

Step 2: L_2 passes through $(6, 0)$ (intersects x -axis at $x = 6$).

$$0 = -\frac{3}{2}(6) + c \implies c = 9$$

On the x -axis $y = 0$. Substitute $(6, 0)$: $-\frac{3}{2} \times 6 = -9$, so $c = 9$.

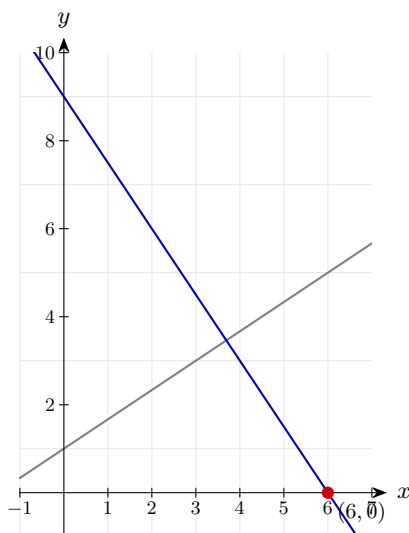
Step 3: Write the equation.

$$y = -\frac{3}{2}x + 9$$

Combine the gradient and intercept.

Final answer: $y = -\frac{3}{2}x + 9$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 23 (cg3-023)

Difficulty: Foundation / Marks: 5

Find the equation of the perpendicular bisector of the line segment joining $A(2, 6)$ and $B(8, 2)$. Give your answer in the form $y = mx + c$.

Worked solution.

Step 1: Find the midpoint of AB .

$$M = \left(\frac{2+8}{2}, \frac{6+2}{2} \right) = (5, 4)$$

The perpendicular bisector must pass through the midpoint of AB . Average the coordinates.

Step 2: Find the gradient of AB .

$$m_{AB} = \frac{2-6}{8-2} = \frac{-4}{6} = -\frac{2}{3}$$

Apply the gradient formula. Simplify $-\frac{4}{6}$ by cancelling a factor of 2.

Step 3: Perpendicular gradient:

$$m_{\perp} = \frac{3}{2}$$

Negative reciprocal of $-\frac{2}{3}$ is $\frac{3}{2}$ (flip and change sign — two negatives cancel).

Step 4: Substitute midpoint $(5, 4)$ into $y = \frac{3}{2}x + c$.

$$4 = \frac{3}{2}(5) + c \implies c = 4 - \frac{15}{2} = -\frac{7}{2}$$

$\frac{3}{2} \times 5 = \frac{15}{2}$. Then $4 - \frac{15}{2} = \frac{8}{2} - \frac{15}{2} = -\frac{7}{2}$.

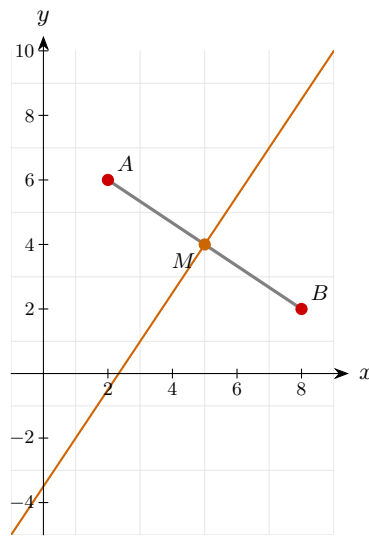
Step 5: Write the equation.

$$y = \frac{3}{2}x - \frac{7}{2}$$

Combine gradient and y -intercept.

Final answer: $y = \frac{3}{2}x - \frac{7}{2}$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 24 (cg3-024)

Difficulty: Foundation / Marks: 5

Find the equation of the perpendicular bisector of the segment joining $P(-2, 1)$ and $Q(4, 7)$. Give your answer in the form $ax + by + c = 0$.

Worked solution.

Step 1: Midpoint of PQ:

$$M = \left(\frac{-2 + 4}{2}, \frac{1 + 7}{2} \right) = (1, 4)$$

Average the x - and y -coordinates. The perpendicular bisector must pass through this point.

Step 2: Gradient of PQ:

$$m_{PQ} = \frac{7 - 1}{4 - (-2)} = \frac{6}{6} = 1$$

Watch the double negative: $4 - (-2) = 6$, not 2.

Step 3: Perpendicular gradient: $m_{\perp} = -1$.

$$y = -x + c$$

Negative reciprocal of 1 is -1 .

Step 4: Substitute (1, 4).

$$4 = -1 + c \implies c = 5 \implies y = -x + 5$$

Substitute the midpoint to find c : $c = 4 + 1 = 5$.

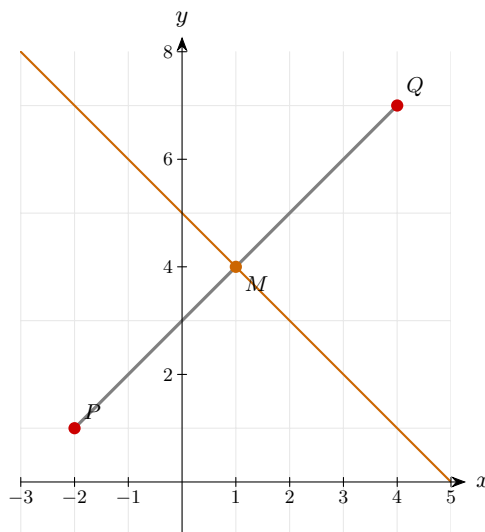
Step 5: Rearrange.

$$x + y - 5 = 0$$

Add x and subtract 5 from both sides to put it in the required form.

Final answer: $x + y - 5 = 0$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 25 (cg3-025)

Difficulty: Foundation / Marks: 6

The perpendicular bisector of segment CD has equation $y = 2x - 3$. If $C = (1, 4)$, find the coordinates of D .

Worked solution.

Step 1: The perpendicular bisector has gradient 2, so CD has gradient $-\frac{1}{2}$.

$$m_{CD} = -\frac{1}{2}$$

CD is perpendicular to its bisector, so its gradient is the negative reciprocal of 2.

Step 2: The midpoint of CD lies on $y = 2x - 3$. Let midpoint be $M(h, k)$ with $k = 2h - 3$.

$$k = 2h - 3$$

Every perpendicular bisector passes through the midpoint of its segment, so M lies on the given line.

Step 3: The gradient from $C(1, 4)$ to $M(h, k)$ equals $-\frac{1}{2}$:

$$\frac{k - 4}{h - 1} = -\frac{1}{2} \implies 2(k - 4) = -(h - 1)$$

Cross-multiplying eliminates the fraction. This second equation lets us pin down M together with the first one.

Step 4: Substitute $k = 2h - 3$:

$$2(2h - 3 - 4) = -(h - 1) \implies 2(2h - 7) = -h + 1 \implies 4h - 14 = -h + 1 \implies 5h = 15 \implies h = 3, k = 3$$

Substitute the line equation into the gradient equation to eliminate k , expand both sides, then solve the linear equation for h . Then $k = 2(3) - 3 = 3$. Midpoint $M = (3, 3)$.

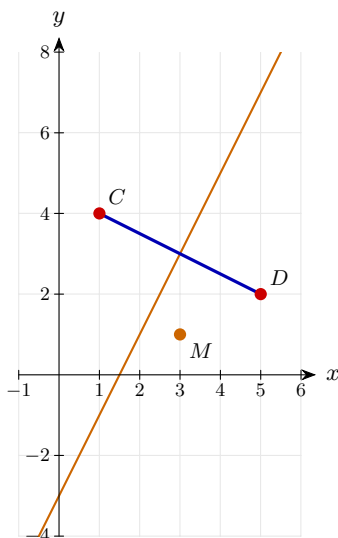
Step 5: Find D : $D = 2M - C$.

$$D = (2(3) - 1, 2(3) - 4) = (5, 2)$$

Since M is the midpoint of CD , $M = \frac{1}{2}(C + D)$, so $D = 2M - C$ coordinate-wise.

Final answer: $D = (5, 2)$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 26 (cg3-026)

Difficulty: Foundation / Marks: 5

Find the equation of the perpendicular bisector of the segment joining $A(0, -3)$ and $B(6, 5)$. Give your answer in the form $ax + by + c = 0$.

Worked solution.

Step 1: Midpoint:

$$M = (3, 1)$$

Averaging coordinates: $\frac{0+6}{2} = 3$ and $\frac{-3+5}{2} = 1$.

Step 2: Gradient of AB :

$$m_{AB} = \frac{5 - (-3)}{6 - 0} = \frac{8}{6} = \frac{4}{3}$$

Watch the double negative: $5 - (-3) = 8$. Then simplify $\frac{8}{6}$ by cancelling a factor of 2.

Step 3: Perpendicular gradient:

$$m_{\perp} = -\frac{3}{4}$$

Negative reciprocal of $\frac{4}{3}$: flip the fraction and change sign.

Step 4: Substitute $M(3, 1)$ into $y = -\frac{3}{4}x + c$:

$$1 = -\frac{9}{4} + c \implies c = \frac{13}{4}$$

$-\frac{3}{4} \times 3 = -\frac{9}{4}$. Then $c = 1 + \frac{9}{4} = \frac{4}{4} + \frac{9}{4} = \frac{13}{4}$.

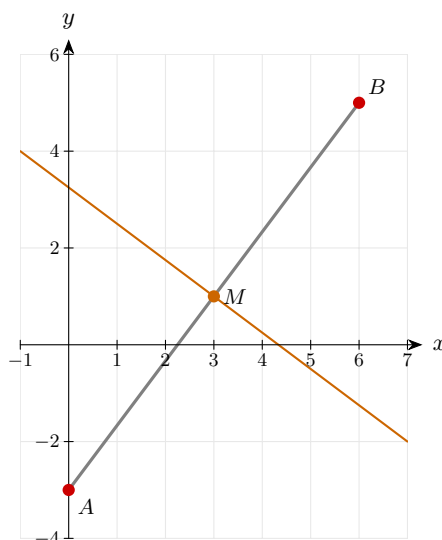
Step 5: Multiply through by 4 and rearrange:

$$4y = -3x + 13 \implies 3x + 4y - 13 = 0$$

Multiplying by 4 clears both fractions in one stroke. Then move $-3x + 13$ to the left.

Final answer: $3x + 4y - 13 = 0$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 27 (cg3-027)

Difficulty: Foundation / Marks: 7

Triangle PQR has vertices $P(0, 4)$, $Q(4, 0)$ and $R(6, 2)$.

(a) Find the equations of sides PQ and QR in the form $y = mx + c$.

(b) Show that PQ and QR are perpendicular.

Worked solution.

Step 1: (a) Gradient of PQ :

$$m_{PQ} = \frac{0 - 4}{4 - 0} = -1; \quad y = -x + 4$$

Use $m = \frac{y_2 - y_1}{x_2 - x_1}$. Since $P(0, 4)$ gives the y -intercept directly, the equation is $y = -x + 4$.

Step 2: Gradient of QR :

$$m_{QR} = \frac{2 - 0}{6 - 4} = 1; \quad y - 0 = 1(x - 4) \implies y = x - 4$$

Same gradient formula, then use point-slope through $Q(4, 0)$ to find the equation.

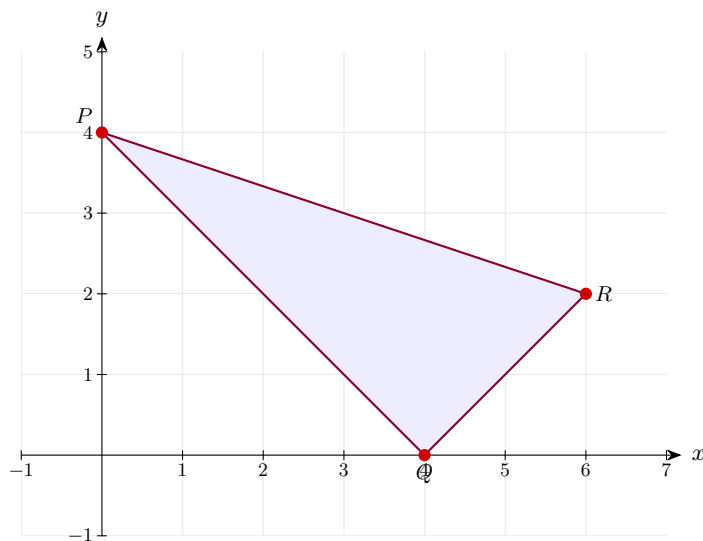
Step 3: (b) Product of gradients:

$$(-1) \times 1 = -1 \implies PQ \perp QR \checkmark$$

Two non-vertical lines are perpendicular if and only if the product of their gradients is -1 .

Final answer: (a) $PQ : y = -x + 4$; $QR : y = x - 4$; (b) Shown — product of gradients $= -1$.

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 28 (cg3-028)

Difficulty: Foundation / Marks: 8

Triangle ABC has vertices $A(0, 0)$, $B(4, 2)$ and $C(2, 6)$.

(a) Find the equations of AB , BC and AC in the form $y = mx + c$.

(b) What type of triangle is ABC ? Justify your answer.

Worked solution.

Step 1: (a) Gradient of AB:

$$m_{AB} = \frac{2 - 0}{4 - 0} = \frac{1}{2}; \quad y = \frac{1}{2}x$$

Since A is the origin, the line through A with gradient $\frac{1}{2}$ has zero y -intercept.

Step 2: Gradient of BC:

$$m_{BC} = \frac{6 - 2}{2 - 4} = \frac{4}{-2} = -2; \quad y = -2x + 10$$

Use point-slope through $B(4, 2)$: $y - 2 = -2(x - 4)$. Check: at $x = 4$, $y = -8 + 10 = 2$ ✓.

Step 3: Gradient of AC:

$$m_{AC} = \frac{6 - 0}{2 - 0} = 3; \quad y = 3x$$

A is the origin, so again the line through A is $y = 3x$.

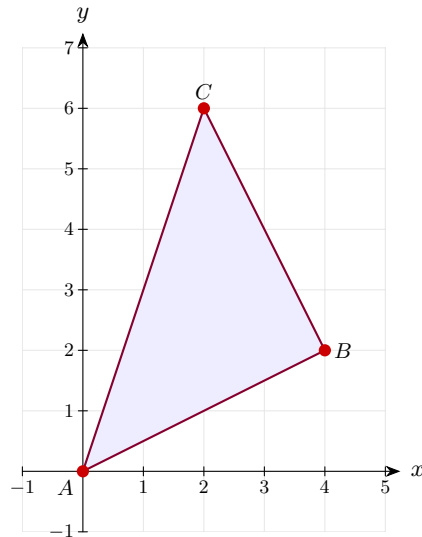
Step 4: (b) Check perpendicularity: $m_{AB} \times m_{BC} = \frac{1}{2} \times (-2) = -1$.

$$m_{AB} \times m_{BC} = -1 \implies AB \perp BC$$

Triangle ABC is right-angled at B .

Final answer: (a) $AB : y = \frac{1}{2}x$; $BC : y = -2x + 10$; $AC : y = 3x$; (b) Right-angled at B (since $AB \perp BC$).

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 29 (cg3-029)

Difficulty: Foundation / Marks: 6

Point $A = (2, 5)$. The line L has equation $y = 3x - 4$. Find the equation of the line through A perpendicular to L , and find where it meets L .

Worked solution.

Step 1: Perpendicular gradient: $m_{\perp} = -\frac{1}{3}$.

$$y - 5 = -\frac{1}{3}(x - 2) \implies y = -\frac{1}{3}x + \frac{17}{3}$$

Negative reciprocal of 3. Apply point-slope form through $A(2, 5)$; expanding gives $y = -\frac{1}{3}x + \frac{2}{3} + 5 = -\frac{1}{3}x + \frac{17}{3}$.

Step 2: Find intersection with $y = 3x - 4$:

$$3x - 4 = -\frac{1}{3}x + \frac{17}{3}$$

Set the two y -expressions equal — at the intersection point both lines share the same (x, y) .

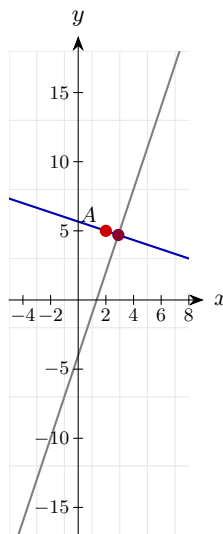
Step 3: Multiply through by 3:

$$9x - 12 = -x + 17 \implies 10x = 29 \implies x = 2.9, y = 3(2.9) - 4 = 4.7$$

Multiplying by 3 clears the fractions. Then collect x -terms and substitute back into either line to find y .

Final answer: Perpendicular: $y = -\frac{1}{3}x + \frac{17}{3}$; meets L at $(2.9, 4.7)$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 30 (cg3-030)

Difficulty: Foundation / Marks: 6

Two lines are given: $L_1 : 2x - y + 4 = 0$ and $L_2 : x + 2y - 3 = 0$.

(a) Show that L_1 and L_2 are perpendicular.

(b) Find their point of intersection.

Worked solution.

Step 1: (a) Rearrange each line.

$$L_1 : y = 2x + 4 \implies m_1 = 2; \quad L_2 : y = -\frac{1}{2}x + \frac{3}{2} \implies m_2 = -\frac{1}{2}$$

Isolate y in each equation to read off gradients.

Step 2: Product of gradients:

$$2 \times \left(-\frac{1}{2}\right) = -1 \implies L_1 \perp L_2 \checkmark$$

Two non-vertical lines are perpendicular if and only if the product of gradients is -1 .

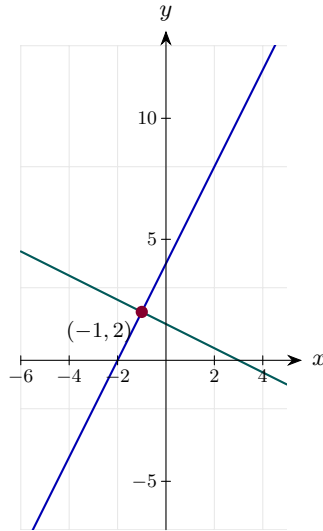
Step 3: (b) Solve simultaneously. From L_2 : $x = 3 - 2y$. Substitute into L_1 :

$$2(3 - 2y) - y + 4 = 0 \implies 6 - 4y - y + 4 = 0 \implies 5y = 10 \implies y = 2, \quad x = -1$$

Substitution eliminates x . Expanding gives $10 - 5y = 0$, so $y = 2$. Then $x = 3 - 4 = -1$.

Final answer: (a) Shown — gradients multiply to -1 ; (b) Intersection at $(-1, 2)$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 31 (cg3-031)

Difficulty: Foundation / Marks: 8

Line L_1 passes through $A(1, 3)$ and $B(5, 11)$.

(a) Find the equation of L_1 in the form $y = mx + c$.

(b) Line L_2 is parallel to L_1 and passes through $C(3, -2)$. Find the equation of L_2 .

(c) Line L_3 is perpendicular to L_1 and passes through B . Find the equation of L_3 .

Worked solution.

Step 1: (a) Gradient of L_1 :

$$m = \frac{11 - 3}{5 - 1} = 2; \quad 3 = 2(1) + c \implies c = 1 \implies y = 2x + 1$$

Apply the gradient formula to A and B . Then substitute $A(1, 3)$ into $y = 2x + c$ to find the intercept.

Step 2: (b) L_2 has gradient 2, passes through $C(3, -2)$:

$$-2 = 2(3) + c \implies c = -8 \implies y = 2x - 8$$

Parallel lines share gradients. Substitute C : $c = -2 - 6 = -8$.

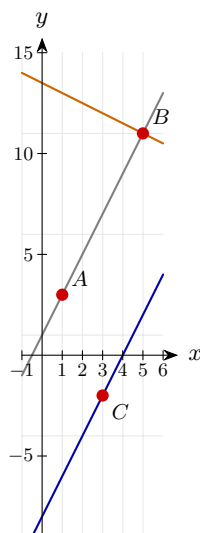
Step 3: (c) L_3 has gradient $-\frac{1}{2}$, passes through $B(5, 11)$:

$$11 = -\frac{1}{2}(5) + c \implies c = \frac{27}{2} \implies y = -\frac{1}{2}x + \frac{27}{2}$$

Perpendicular gradient is the negative reciprocal of 2. Then $c = 11 + \frac{5}{2} = \frac{22}{2} + \frac{5}{2} = \frac{27}{2}$.

Final answer: (a) $y = 2x + 1$; (b) $y = 2x - 8$; (c) $y = -\frac{1}{2}x + \frac{27}{2}$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 32 (cg3-032)

Difficulty: Foundation / Marks: 8

The line L has equation $3x + 4y - 24 = 0$.

(a) Find where L crosses the axes.

(b) Find the equation of the line parallel to L that passes through $(4, -1)$. Give your answer in the form $ax + by + c = 0$.

(c) Find the equation of the line perpendicular to L that passes through $(0, 0)$. Give your answer in the form $y = mx$.

Worked solution.

Step 1: (a) x -intercept (set $y = 0$): $3x = 24 \implies x = 8$. y -intercept (set $x = 0$): $4y = 24 \implies y = 6$.

x -intercept : $(8, 0)$; y -intercept : $(0, 6)$

Setting one variable to zero gives the intercept on the other axis directly.

Step 2: (b) Gradient of L : $m = -\frac{3}{4}$. Parallel line through $(4, -1)$:

$$-1 = -\frac{3}{4}(4) + c \implies c = 2 \implies y = -\frac{3}{4}x + 2 \implies 3x + 4y - 8 = 0$$

From $4y = -3x + 24$, the gradient is $-\frac{3}{4}$. Substitute the point, then multiply by 4 and rearrange to integer form.

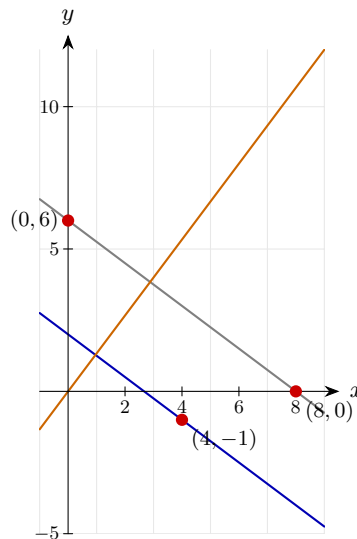
Step 3: (c) Perpendicular gradient: $m_{\perp} = \frac{4}{3}$. Through $(0, 0)$:

$$y = \frac{4}{3}x$$

Through the origin, $c = 0$, giving the required form $y = mx$ directly.

Final answer: (a) $(8, 0)$ and $(0, 6)$; (b) $3x + 4y - 8 = 0$; (c) $y = \frac{4}{3}x$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 33 (cg3-033)

Difficulty: Foundation / Marks: 8

The vertices of a quadrilateral are $A(0, 2)$, $B(4, 4)$, $C(6, 0)$ and $D(2, -2)$.

(a) Find the gradients of all four sides.

(b) Show that $ABCD$ is a rectangle.

Worked solution.

Step 1: (a) Gradient of each side:

$$m_{AB} = \frac{4 - 2}{4 - 0} = \frac{1}{2}, \quad m_{BC} = \frac{0 - 4}{6 - 4} = -2$$

Apply the gradient formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ to each pair of vertices in turn.

$$m_{CD} = \frac{-2 - 0}{2 - 6} = \frac{-2}{-4} = \frac{1}{2}, \quad m_{DA} = \frac{2 - (-2)}{0 - 2} = -2$$

Watch the double negatives. For m_{CD} both numerator and denominator are negative, giving a positive result. For m_{DA} , $2 - (-2) = 4$.

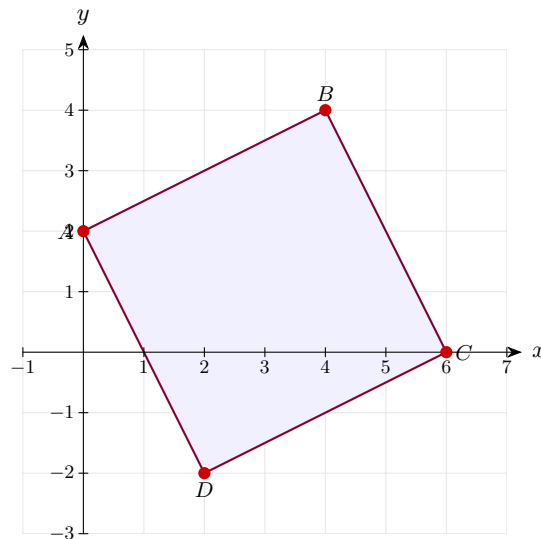
Step 3: (b) $AB \parallel CD$ (both gradient $\frac{1}{2}$) and $BC \parallel DA$ (both gradient -2), so $ABCD$ is a parallelogram.

$$m_{AB} \times m_{BC} = \frac{1}{2} \times (-2) = -1$$

Adjacent sides are perpendicular, so the parallelogram is a rectangle. ✓

Final answer: (a) $m_{AB} = m_{CD} = \frac{1}{2}$, $m_{BC} = m_{DA} = -2$; (b) Opposite sides parallel and adjacent sides perpendicular $\Rightarrow$ rectangle.

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 34 (cg3-034)

Difficulty: Foundation / Marks: 9

Line L_1 has equation $y = \frac{1}{2}x + 3$. Line L_2 is perpendicular to L_1 and passes through $(4, 1)$.

- Find the equation of L_2 in the form $ax + by + c = 0$.
- Find the coordinates of the point where L_1 and L_2 intersect.
- Find the exact distance between the point $(4, 1)$ and the intersection point.

Worked solution.

Step 1: (a) Perpendicular gradient: $m_{\perp} = -2$. Through $(4, 1)$:

$$1 = -2(4) + c \implies c = 9 \implies y = -2x + 9 \implies 2x + y - 9 = 0$$

Negative reciprocal of $\frac{1}{2}$ is -2 . Substitute the given point to find c , then rearrange to the required form.

Step 2: (b) Solve $y = \frac{1}{2}x + 3$ and $y = -2x + 9$ simultaneously:

$$\frac{1}{2}x + 3 = -2x + 9 \implies \frac{5}{2}x = 6 \implies x = \frac{12}{5}, y = -2\left(\frac{12}{5}\right) + 9 = \frac{21}{5}$$

Set the y -expressions equal. Collecting x -terms: $\frac{1}{2}x + 2x = \frac{5}{2}x$. Then divide by $\frac{5}{2}$, i.e. multiply by $\frac{2}{5}$.

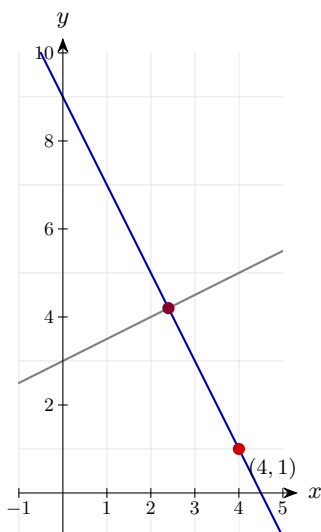
Step 3: (c) Distance from $(4, 1)$ to $(\frac{12}{5}, \frac{21}{5})$:

$$d = \sqrt{\left(4 - \frac{12}{5}\right)^2 + \left(1 - \frac{21}{5}\right)^2} = \sqrt{\left(\frac{8}{5}\right)^2 + \left(-\frac{16}{5}\right)^2} = \sqrt{\frac{64 + 256}{25}} = \sqrt{\frac{320}{25}} = \frac{8\sqrt{5}}{5}$$

Apply the distance formula (Pythagoras on the coordinate differences). Convert to common fifths: $4 = \frac{20}{5}$ and $1 = \frac{5}{5}$. $\sqrt{320} = \sqrt{64 \times 5} = 8\sqrt{5}$.

Final answer: (a) $2x + y - 9 = 0$; (b) $\left(\frac{12}{5}, \frac{21}{5}\right)$; (c) $\frac{8\sqrt{5}}{5}$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 35 (cg3-035)

Difficulty: Foundation / Marks: 9

Points $A(2, 7)$, $B(8, 5)$ and $C(k, 1)$ are given.

- (a) Find the equation of the perpendicular bisector of AB .
- (b) Given that C lies on the perpendicular bisector of AB , find the value of k .
- (c) Find the exact distance AC .

Worked solution.

Step 1: (a) Midpoint of AB :

$$M = \left(\frac{2+8}{2}, \frac{7+5}{2} \right) = (5, 6)$$

Average the endpoint coordinates. The perpendicular bisector must pass through this midpoint.

Step 2: Gradient of AB :

$$m_{AB} = \frac{5-7}{8-2} = -\frac{1}{3} \implies m_{\perp} = 3$$

Use the gradient formula, then take the negative reciprocal: flip $-\frac{1}{3}$ and change sign to get 3.

Step 3: Perpendicular bisector through $M(5, 6)$ with gradient 3:

$$6 = 3(5) + c \implies c = -9 \implies y = 3x - 9$$

Substitute the midpoint to find c : $c = 6 - 15 = -9$.

Step 4: (b) Substitute $C(k, 1)$ into $y = 3x - 9$:

$$1 = 3k - 9 \implies 3k = 10 \implies k = \frac{10}{3}$$

Since C lies on the bisector its coordinates satisfy the equation. Solve the resulting linear equation in k .

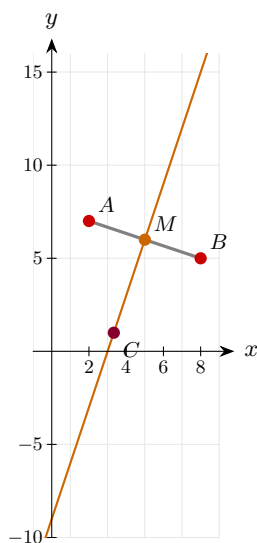
Step 5: (c) $A = (2, 7)$, $C = (\frac{10}{3}, 1)$:

$$AC = \sqrt{\left(\frac{10}{3} - 2\right)^2 + (1 - 7)^2} = \sqrt{\left(\frac{4}{3}\right)^2 + 36} = \sqrt{\frac{16}{9} + \frac{324}{9}} = \sqrt{\frac{340}{9}} = \frac{2\sqrt{85}}{3}$$

Distance formula. Convert 36 to ninths so the surds combine: $36 = \frac{324}{9}$. Finally $\sqrt{340} = \sqrt{4 \times 85} = 2\sqrt{85}$, and $\sqrt{9} = 3$.

Final answer: (a) $y = 3x - 9$; (b) $k = \frac{10}{3}$; (c) $AC = \frac{2\sqrt{85}}{3}$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 36 (cg3-036)

Difficulty: Foundation / Marks: 2

Find the equation of the line parallel to $y = 4x - 1$ that passes through $(2, 3)$.

Worked solution.

Step 1: Same gradient

$$m = 4$$

Parallel lines share gradients, so the new line also has gradient 4.

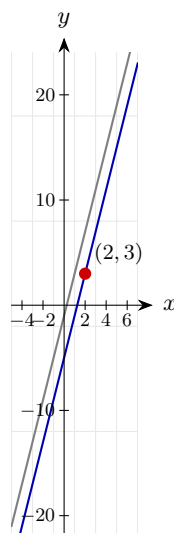
Step 2: Point-slope

$$y - 3 = 4(x - 2) \implies y = 4x - 5$$

Apply $y - y_1 = m(x - x_1)$ through $(2, 3)$, then expand and simplify.

Final answer: $y = 4x - 5$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 37 (cg3-037)

Difficulty: Foundation / Marks: 3

Find the equation of the line perpendicular to $y = 3x + 2$ passing through $(6, 1)$.

Worked solution.

Step 1: Perpendicular gradient

$$m = -\frac{1}{3}$$

Negative reciprocal of 3.

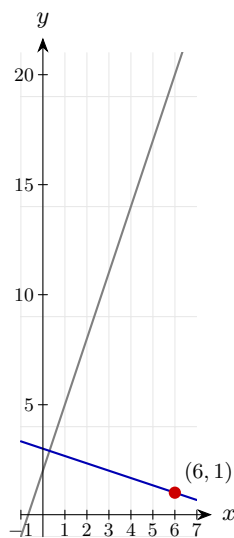
Step 2: Equation

$$y - 1 = -\frac{1}{3}(x - 6) \implies y = -\frac{1}{3}x + 3$$

Expand: $-\frac{1}{3}(x - 6) = -\frac{1}{3}x + 2$, then add 1 to both sides.

Final answer: $y = -\frac{1}{3}x + 3$
--

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 38 (cg3-038)

Difficulty: Foundation / Marks: 2

Determine whether the lines $y = \frac{2}{5}x + 1$ and $2x - 5y + 10 = 0$ are parallel, perpendicular, or neither.

Worked solution.

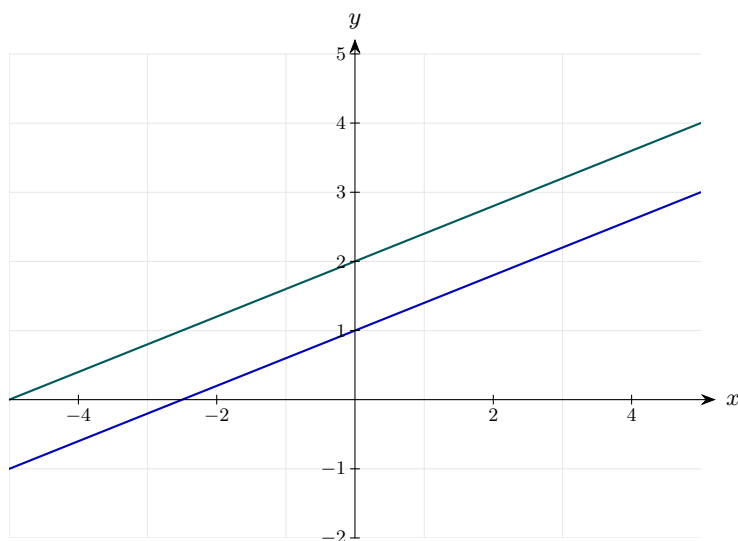
Find gradients

$$m_1 = \frac{2}{5}; \quad 5y = 2x + 10 \implies m_2 = \frac{2}{5}$$

Equal gradients.

Final answer: Parallel

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 39 (cg3-039)

Difficulty: Foundation / Marks: 2

Find the value of k if the lines $y = kx + 3$ and $y = -2x + 5$ are perpendicular.

Worked solution.

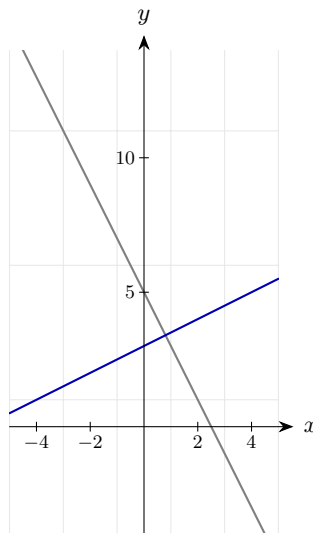
Product of gradients = -1

$$k \times (-2) = -1 \implies k = \frac{1}{2}$$

Perpendicular lines have gradients whose product is -1 . Dividing both sides by -2 gives $k = \frac{1}{2}$.

Final answer: $k = \frac{1}{2}$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 40 (cg3-040)

Difficulty: Foundation / Marks: 2

The line L_1 has equation $3x + 4y = 12$. Find the equation of the line parallel to L_1 passing through the origin.

Worked solution.

Step 1: Gradient of L_1

$$4y = -3x + 12 \implies m = -\frac{3}{4}$$

Isolate y by subtracting $3x$ and dividing by 4. Coefficient of x is the gradient.

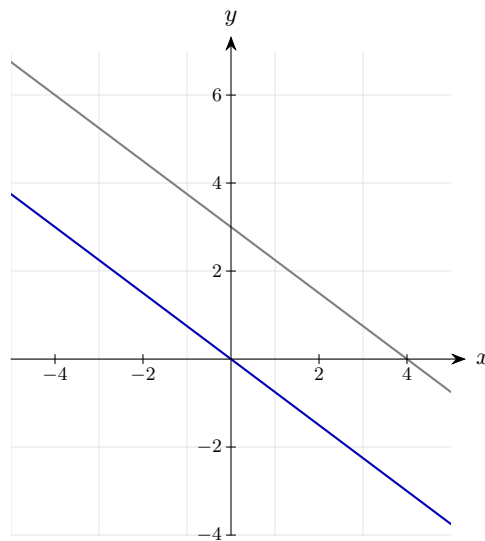
Step 2: Through origin

$$y = -\frac{3}{4}x$$

Through the origin, the y -intercept is 0, so $c = 0$.

Final answer: $y = -\frac{3}{4}x$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 41 (cg3-041)

Difficulty: Foundation / Marks: 3

Find the equation of the line perpendicular to $2x - 5y + 10 = 0$ passing through $(4, -1)$.

Worked solution.

Step 1: Gradient of given line

$$5y = 2x + 10 \implies m = \frac{2}{5}$$

Move $-2x - 10$ to the right and divide by 5 to read off the gradient.

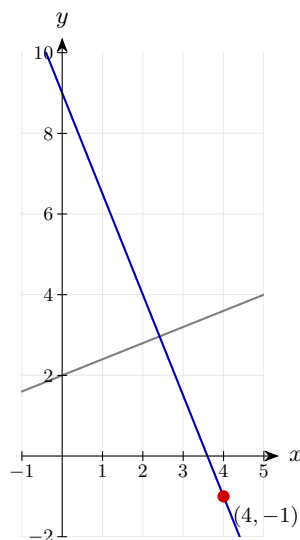
Step 2: Perpendicular gradient and equation

$$m_{\perp} = -\frac{5}{2}; \quad y + 1 = -\frac{5}{2}(x - 4) \implies y = -\frac{5}{2}x + 9$$

Negative reciprocal of $\frac{2}{5}$ is $-\frac{5}{2}$. Expand: $-\frac{5}{2}(x - 4) = -\frac{5}{2}x + 10$, then subtract 1.

Final answer: $y = -\frac{5}{2}x + 9$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 42 (cg3-042)

Difficulty: Foundation / Marks: 2

Show that the lines $y = 5x - 3$ and $x + 5y = 20$ are perpendicular.

Worked solution.

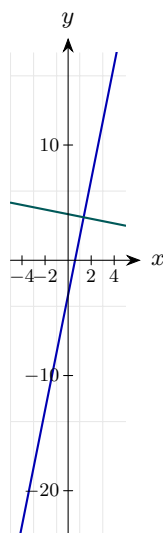
Find gradients and check product

$$m_1 = 5; \quad 5y = -x + 20 \implies m_2 = -\frac{1}{5}; \quad 5 \times \left(-\frac{1}{5}\right) = -1 \checkmark$$

The first line is already in $y = mx + c$ form. Rearrange the second by dividing by 5. The product of the gradients is -1 , confirming perpendicularity.

Final answer: Product of gradients is -1 , so perpendicular.

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 43 (cg3-043)

Difficulty: Foundation / Marks: 3

Find the value of p such that the lines $px + 3y = 6$ and $2x - y = 4$ are parallel.

Worked solution.

Step 1: Gradients

$$m_1 = -\frac{p}{3}; \quad m_2 = 2$$

From $px + 3y = 6$: $3y = -px + 6$, so $m_1 = -\frac{p}{3}$. From $2x - y = 4$: $y = 2x - 4$, so $m_2 = 2$.

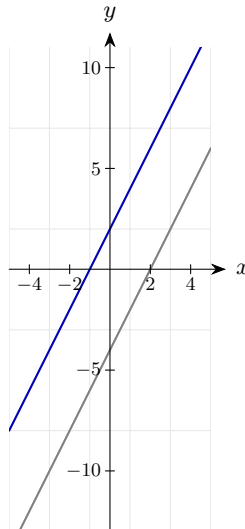
Step 2: Set equal for parallel

$$-\frac{p}{3} = 2 \implies p = -6$$

Multiply both sides by -3 ; the sign flips, giving $p = -6$.

Final answer: $p = -6$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 44 (cg3-044)

Difficulty: Foundation / Marks: 4

The line L passes through $A(1, 5)$ and $B(3, -1)$. Find the equation of the line through $(4, 2)$ perpendicular to L .

Worked solution.

Step 1: Gradient of L

$$m_L = \frac{-1 - 5}{3 - 1} = -3$$

Apply the gradient formula to A and B : $\frac{-6}{2} = -3$.

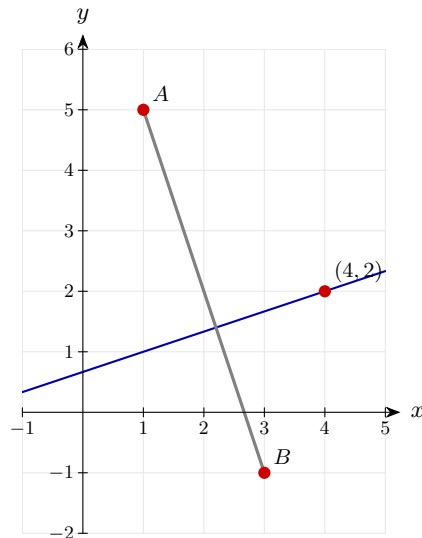
Step 2: Perpendicular and equation

$$m_{\perp} = \frac{1}{3}; \quad y - 2 = \frac{1}{3}(x - 4) \implies y = \frac{1}{3}x + \frac{2}{3}$$

Negative reciprocal of -3 is $\frac{1}{3}$. Expand: $\frac{1}{3}(x - 4) = \frac{1}{3}x - \frac{4}{3}$, then add $2 = \frac{6}{3}$ to get $+\frac{2}{3}$.

Final answer: $y = \frac{1}{3}x + \frac{2}{3}$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 45 (cg3-045)

Difficulty: Foundation / Marks: 5

Find the foot of the perpendicular from $P(3, 7)$ to the line $y = 2x + 1$.

Worked solution.

Step 1: Perpendicular from P has gradient $-1/2$

$$y - 7 = -\frac{1}{2}(x - 3) \implies y = -\frac{1}{2}x + \frac{17}{2}$$

The perpendicular gradient is the negative reciprocal of 2. Use point-slope form through $P(3, 7)$.

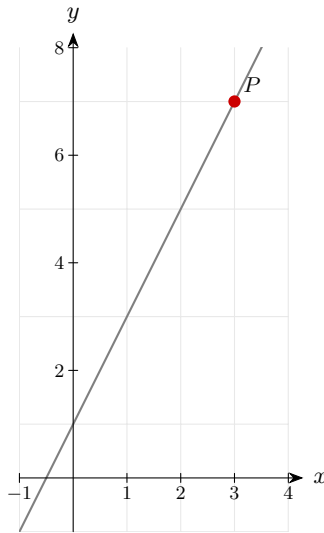
Step 2: Solve simultaneously

$$2x + 1 = -\frac{1}{2}x + \frac{17}{2} \implies \frac{5}{2}x = \frac{15}{2} \implies x = 3, y = 7$$

Setting the two equations equal finds their intersection. Quick check: $y = 2(3) + 1 = 7$, so P itself lies on the line — the foot of the perpendicular from a point on the line is that point.

Final answer: $P(3, 7)$ lies on the line, so the foot of the perpendicular is P itself.

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 46 (cg3-046)

Difficulty: Foundation / Marks: 4

Find the foot of the perpendicular from $P(4, 5)$ to the line $y = x + 1$.

Worked solution.

Step 1: Perpendicular from P

$$m_{\perp} = -1; \quad y - 5 = -(x - 4) \implies y = -x + 9$$

The line $y = x + 1$ has gradient 1, so the perpendicular has gradient -1 . Point-slope form through $P(4, 5)$.

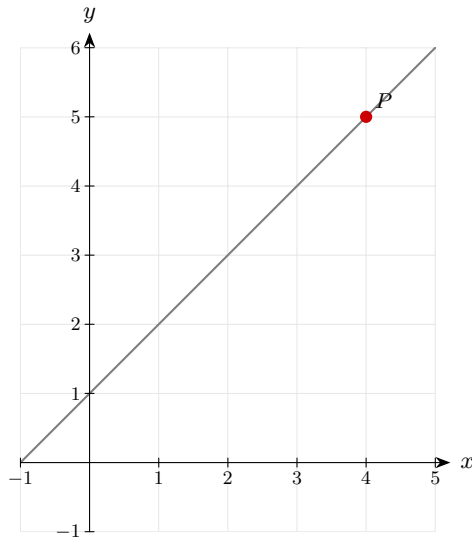
Step 2: Solve with $y = x + 1$

$$x + 1 = -x + 9 \implies 2x = 8 \implies x = 4, \quad y = 5$$

Setting the equations equal gives the foot of the perpendicular. Note that $P(4, 5)$ satisfies $y = x + 1$, so P lies on the line and the foot is P itself.

Final answer: $(4, 5)$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 47 (cg3-047)

Difficulty: Foundation / Marks: 4

Find the foot of the perpendicular from $P(1, 6)$ to the line $y = 3x - 2$.

Worked solution.

Step 1: Perpendicular from P

$$m_{\perp} = -\frac{1}{3}; \quad y - 6 = -\frac{1}{3}(x - 1) \implies y = -\frac{1}{3}x + \frac{19}{3}$$

Negative reciprocal of 3. Expand and add: $\frac{1}{3} + 6 = \frac{1}{3} + \frac{18}{3} = \frac{19}{3}$.

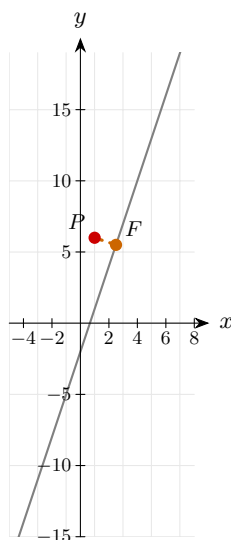
Step 2: Solve simultaneously

$$3x - 2 = -\frac{1}{3}x + \frac{19}{3} \implies \frac{10}{3}x = \frac{25}{3} \implies x = \frac{5}{2}, y = \frac{11}{2}$$

Add $\frac{1}{3}x$ to both sides: $3x + \frac{1}{3}x = \frac{10}{3}x$. Then $\frac{25}{3} \div \frac{10}{3} = \frac{5}{2}$. Substitute back into $y = 3x - 2$ to get $y = \frac{15}{2} - 2 = \frac{11}{2}$.

Final answer: $\left(\frac{5}{2}, \frac{11}{2}\right)$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 48 (cg3-048)

Difficulty: Foundation / Marks: 4

The line $L_1 : y = -\frac{3}{4}x + 6$ meets the x -axis at A . Find the equation of the line through A perpendicular to L_1 .

Worked solution.

Step 1: Find A ($y = 0$)

$$0 = -\frac{3}{4}x + 6 \implies x = 8, \quad A = (8, 0)$$

At the x -axis, $y = 0$. Solve for x : $\frac{3}{4}x = 6$, so $x = 8$.

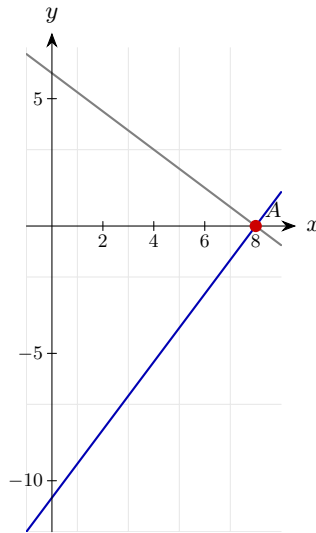
Step 2: Perpendicular gradient and equation

$$m_{\perp} = \frac{4}{3}; \quad y = \frac{4}{3}(x - 8) = \frac{4}{3}x - \frac{32}{3}$$

Negative reciprocal of $-\frac{3}{4}$ is $\frac{4}{3}$. Since the line passes through $A(8, 0)$, use point-slope with $y_1 = 0$.

Final answer: $y = \frac{4}{3}x - \frac{32}{3}$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 49 (cg3-049)

Difficulty: Foundation / Marks: 3

Find the values of k for which $kx + 2y = 5$ is perpendicular to $3x - 4y = 1$.

Worked solution.

Step 1: Gradients

$$m_1 = -\frac{k}{2}; \quad m_2 = \frac{3}{4}$$

From $kx + 2y = 5$: $y = -\frac{k}{2}x + \frac{5}{2}$. From $3x - 4y = 1$: $4y = 3x - 1$, so $m_2 = \frac{3}{4}$.

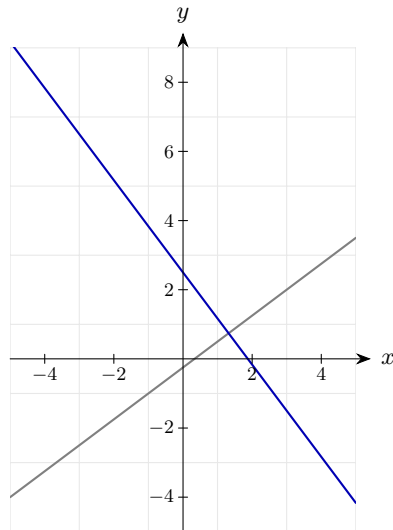
Step 2: Product = -1

$$-\frac{k}{2} \times \frac{3}{4} = -1 \implies \frac{3k}{8} = 1 \implies k = \frac{8}{3}$$

Multiplying gives $-\frac{3k}{8} = -1$, so $\frac{3k}{8} = 1$. Multiply both sides by $\frac{8}{3}$.

Final answer: $k = \frac{8}{3}$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 50 (cg3-050)

Difficulty: Foundation / Marks: 3

A triangle has vertices $A(0,0)$, $B(6,0)$, $C(2,4)$. Find the equation of the altitude from C to AB .

Worked solution.

Step 1: AB is horizontal ($y = 0$)

$$m_{AB} = 0$$

Both A and B have $y = 0$, so AB lies along the x -axis.

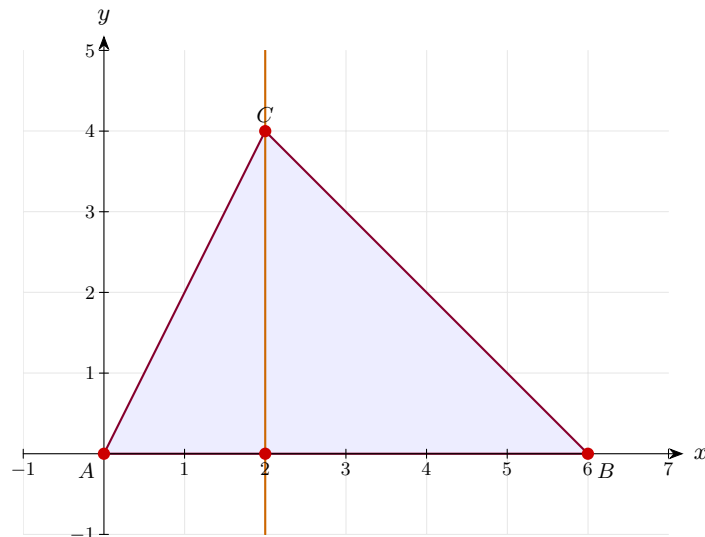
Step 2: Altitude from C is vertical

$$x = 2$$

Perpendicular to a horizontal line is vertical.

Final answer: $x = 2$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 51 (cg3-051)

Difficulty: Foundation / Marks: 4

Find the equation of the altitude from $A(0,0)$ to BC where $B(6,0)$ and $C(2,4)$.

Worked solution.

Step 1: Gradient of BC

$$m_{BC} = \frac{4-0}{2-6} = -1$$

Apply the gradient formula: $\frac{4}{-4} = -1$. The altitude from A is perpendicular to BC .

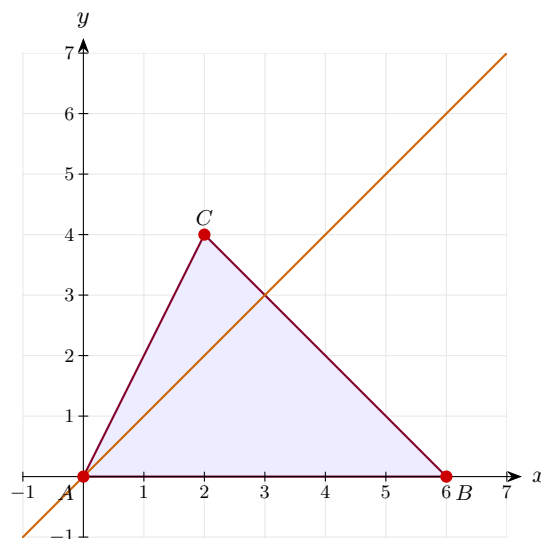
Step 2: Perpendicular gradient and altitude

$$m_{\perp} = 1; \quad y = x$$

Through origin with gradient 1.

Final answer: $y = x$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 52 (cg3-052)

Difficulty: Foundation / Marks: 2

Find the perpendicular distance from $(2, -1)$ to the line $4x + 3y - 7 = 0$.

Worked solution.

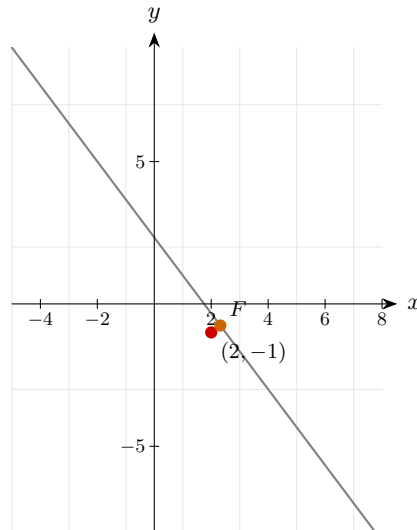
Distance formula

$$d = \frac{|4(2) + 3(-1) - 7|}{\sqrt{16 + 9}} = \frac{|8 - 3 - 7|}{5} = \frac{2}{5}$$

The perpendicular distance from (x_0, y_0) to $ax + by + c = 0$ is $\frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$. The absolute value handles either side of the line.

Final answer: $\frac{2}{5}$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 53 (cg3-053)

Difficulty: Foundation / Marks: 3

The lines $y = 2x + c$ and $y = 2x + d$ are 3 units apart. Find $|c - d|$.

Worked solution.

Step 1: Distance between parallel lines

$$d = \frac{|c - d|}{\sqrt{1 + 4}} = \frac{|c - d|}{\sqrt{5}}$$

Rewrite as $2x - y + c = 0$ and $2x - y + d = 0$.

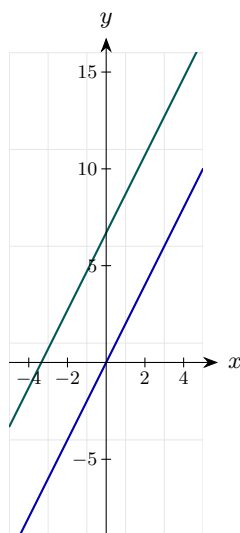
Step 2: Solve

$$\frac{|c - d|}{\sqrt{5}} = 3 \implies |c - d| = 3\sqrt{5}$$

Multiply both sides by $\sqrt{5}$ to isolate $|c - d|$.

Final answer: $|c - d| = 3\sqrt{5}$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 54 (cg3-054)

Difficulty: Foundation / Marks: 3

Find the equation of the line through $(5, -3)$ parallel to $2x + 7y = 14$. Give your answer in the form $ax + by + c = 0$.

Worked solution.

Step 1: Gradient

$$m = -\frac{2}{7}$$

From $2x + 7y = 14$: $7y = -2x + 14$, so the gradient is $-\frac{2}{7}$. Parallel lines share this.

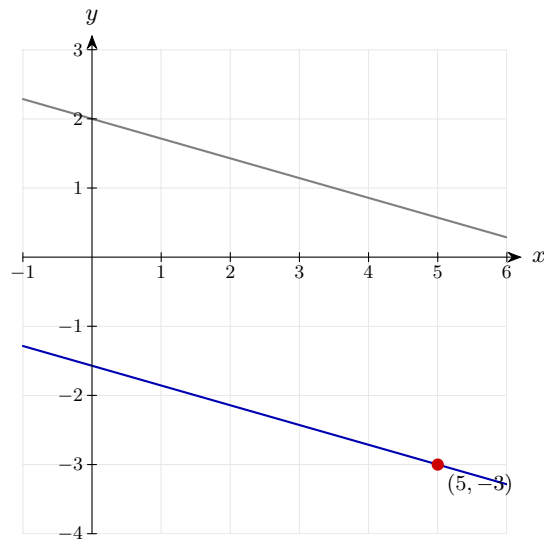
Step 2: Equation

$$y + 3 = -\frac{2}{7}(x - 5) \implies 7y + 21 = -2x + 10 \implies 2x + 7y + 11 = 0$$

Use point-slope through $(5, -3)$, multiply by 7 to clear the fraction, then collect everything on one side.

Final answer: $2x + 7y + 11 = 0$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 55 (cg3-055)

Difficulty: Foundation / Marks: 3

Two perpendicular lines meet at $(3, 4)$. One has gradient 2. Find the equations of both lines.

Worked solution.

Step 1: Line 1

$$y - 4 = 2(x - 3) \implies y = 2x - 2$$

Use point-slope through $(3, 4)$ with gradient 2, then expand: $2(x - 3) = 2x - 6$, then add 4.

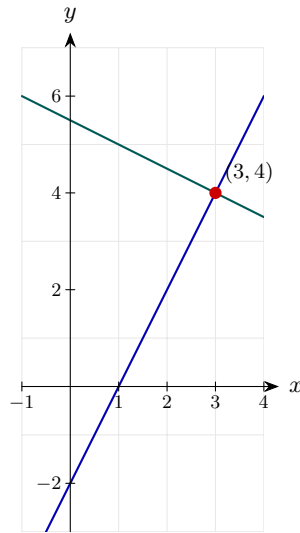
Step 2: Line 2 (perpendicular)

$$y - 4 = -\frac{1}{2}(x - 3) \implies y = -\frac{1}{2}x + \frac{11}{2}$$

Negative reciprocal of 2 is $-\frac{1}{2}$. Expand and add: $\frac{3}{2} + 4 = \frac{3}{2} + \frac{8}{2} = \frac{11}{2}$.

Final answer: $y = 2x - 2$ and $y = -\frac{1}{2}x + \frac{11}{2}$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 56 (cg3-056)

Difficulty: Foundation / Marks: 5

Show that the quadrilateral with vertices $A(0, 1)$, $B(4, 3)$, $C(3, 5)$, $D(-1, 3)$ is a rectangle.

Worked solution.

Step 1: Gradient of AB

$$m_{AB} = \frac{2}{4} = \frac{1}{2}$$

$$m_{AB} = \frac{3-1}{4-0} = \frac{2}{4} = \frac{1}{2}.$$

Step 2: Gradient of BC

$$m_{BC} = \frac{2}{-1} = -2$$

$$m_{BC} = \frac{5-3}{3-4} = \frac{2}{-1} = -2.$$

Step 3: Check perpendicular

$$\frac{1}{2} \times (-2) = -1 \checkmark$$

AB perp BC.

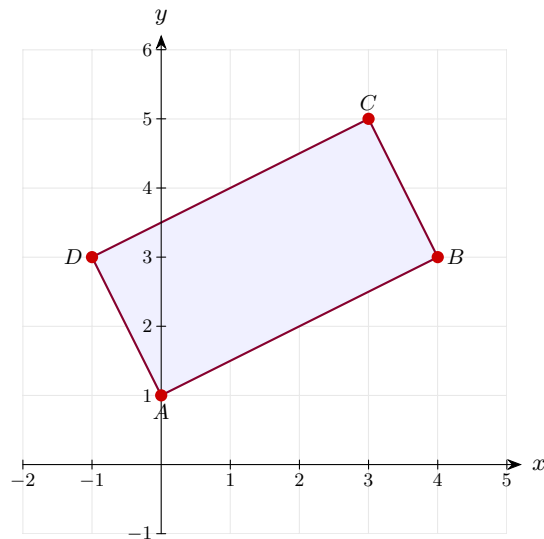
Step 4: Check parallel opposite sides

$$m_{CD} = \frac{-2}{-4} = \frac{1}{2} = m_{AB}; \quad m_{DA} = \frac{-2}{1} = -2 = m_{BC}$$

Opposite sides parallel and adjacent sides perpendicular: rectangle.

Final answer: Opposite sides parallel, adjacent sides perpendicular: ABCD is a rectangle.

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 57 (cg3-057)

Difficulty: Foundation / Marks: 3

The line L passes through $(0, 5)$ perpendicular to $y = -\frac{2}{3}x + 4$. Where does L cross the x -axis?

Worked solution.

Step 1: L has gradient $3/2$

$$y = \frac{3}{2}x + 5$$

Negative reciprocal of $-\frac{2}{3}$ is $\frac{3}{2}$. Since the line passes through $(0, 5)$, $c = 5$ directly.

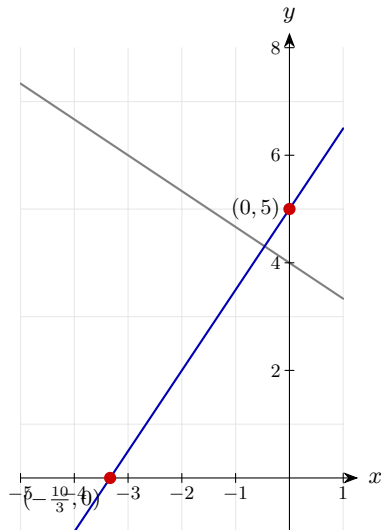
Step 2: Set $y = 0$

$$0 = \frac{3}{2}x + 5 \implies x = -\frac{10}{3}$$

At the x -axis, $y = 0$. Solve $\frac{3}{2}x = -5$, giving $x = -\frac{10}{3}$.

Final answer: $\left(-\frac{10}{3}, 0\right)$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 58 (cg3-058)

Difficulty: Foundation / Marks: 4

Find the equation of the perpendicular bisector of the segment from $(1, -3)$ to $(5, 1)$.

Worked solution.

Step 1: Midpoint

$$M = (3, -1)$$

Average coordinates: $\frac{1+5}{2} = 3$, $\frac{-3+1}{2} = -1$.

Step 2: Gradient of segment

$$m = \frac{4}{4} = 1$$

$$m = \frac{1-(-3)}{5-1} = \frac{4}{4} = 1.$$

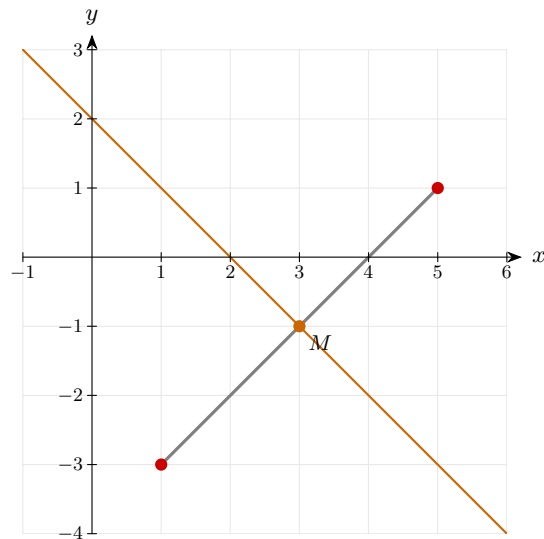
Step 3: Perpendicular bisector

$$m_{\perp} = -1; \quad y + 1 = -(x - 3) \implies y = -x + 2$$

Negative reciprocal of 1 is -1 . Point-slope through the midpoint, then expand: $-(x - 3) = -x + 3$, and subtract 1 from both sides.

Final answer: $y = -x + 2$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 59 (cg3-059)

Difficulty: Foundation / Marks: 2

The line $y = mx + 3$ is parallel to the line joining $(1, 2)$ and $(4, 8)$. Find m .

Worked solution.

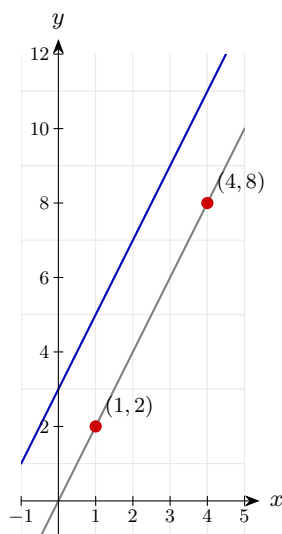
Gradient of line through points

$$m = \frac{8 - 2}{4 - 1} = 2$$

Parallel means same gradient.

Final answer: $m = 2$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 60 (cg3-060)

Difficulty: Foundation / Marks: 3

Three lines have equations $L_1 : y = 3x$, $L_2 : y = -\frac{1}{3}x + 4$, $L_3 : y = 3x - 5$. Which pairs are parallel? Which are perpendicular?

Worked solution.

Step 1: Compare gradients

$$m_1 = 3, m_2 = -\frac{1}{3}, m_3 = 3$$

Read each gradient straight from the $y = mx + c$ form.

Step 2: Parallel

$$L_1 \parallel L_3 \text{ (both gradient 3)}$$

Equal gradients with different intercepts means the lines are parallel.

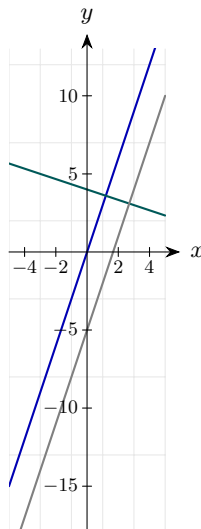
Step 3: Perpendicular

$$L_1 \perp L_2 \text{ and } L_2 \perp L_3 \text{ (product} = -1)$$

$3 \times (-\frac{1}{3}) = -1$, so L_2 is perpendicular to both L_1 and L_3 .

Final answer: L_1 parallel L_3 ; $L_1 \perp L_2$; $L_2 \perp L_3$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 61 (cg3-061)

Difficulty: Foundation / Marks: 2

Find the reflection of the point $(5, 2)$ in the line $y = x$.

Worked solution.

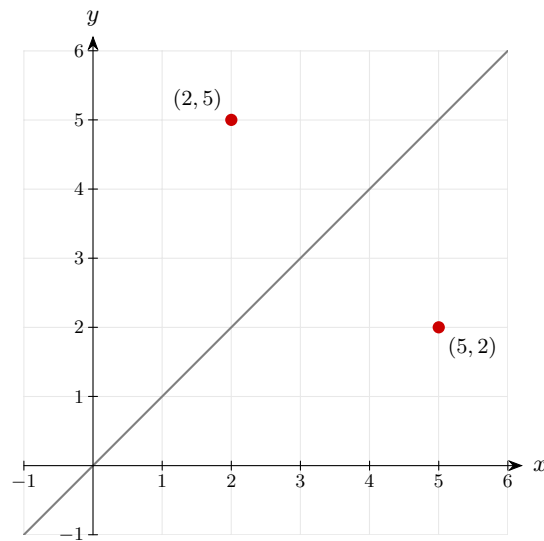
Reflection in $y = x$

$$(5, 2) \rightarrow (2, 5)$$

Swap coordinates.

Final answer: $(2, 5)$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 62 (cg3-062)

Difficulty: Foundation / Marks: 3

Find the value of a if the line $ax + 6y = 12$ is perpendicular to $2x + 3y = 9$.

Worked solution.

Step 1: Gradients

$$m_1 = -\frac{a}{6}; \quad m_2 = -\frac{2}{3}$$

Rearrange each line into $y = mx + c$ form to read off the gradients.

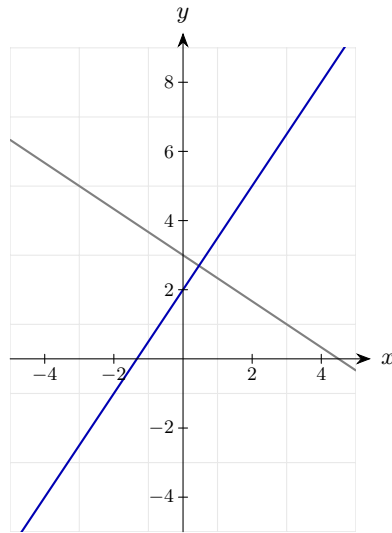
Step 2: Product = -1

$$\left(-\frac{a}{6}\right) \times \left(-\frac{2}{3}\right) = -1 \implies \frac{a}{9} = -1 \implies a = -9$$

Two non-vertical lines are perpendicular if and only if their gradients multiply to -1 . Be careful with the double negative — $(-)(-) = +$ so the product is $+\frac{a}{9}$.

Final answer: $a = -9$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 63 (cg3-063)

Difficulty: Foundation / Marks: 3

A rhombus has vertices at $(0, 3)$, $(4, 0)$, $(0, -3)$, $(-4, 0)$. Show that its diagonals are perpendicular.

Worked solution.

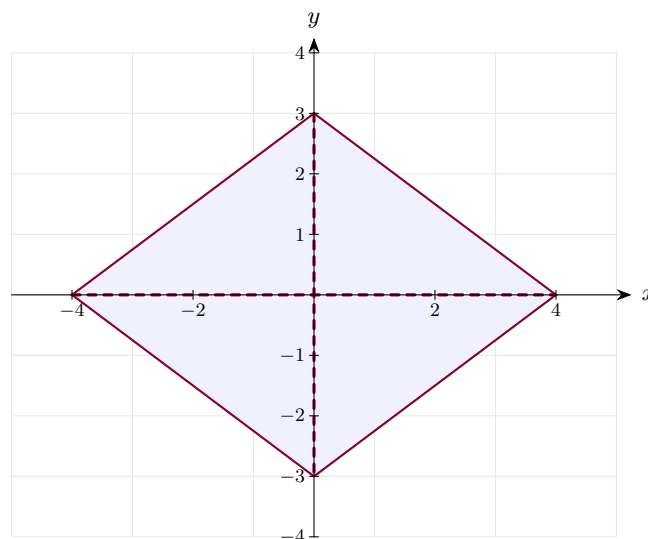
Diagonal gradients

$$m_1 = \frac{3 - (-3)}{0 - 0} = \text{undefined (vertical)}; \quad m_2 = \frac{0 - 0}{4 - (-4)} = 0 \text{ (horizontal)}$$

Vertical and horizontal lines are perpendicular.

Final answer: One diagonal is vertical, the other horizontal: they are perpendicular.

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 64 (cg3-064)

Difficulty: Foundation / Marks: 5

Find the area of the triangle formed by $y = 2x$, $y = -\frac{1}{2}x + 5$, and the x -axis.

Worked solution.

Step 1: Intersection of the two lines

$$2x = -\frac{1}{2}x + 5 \implies \frac{5}{2}x = 5 \implies x = 2, y = 4$$

Equate the two y -expressions to find where the lines meet. This is the apex of the triangle.

Step 2: x -intercepts

$$y = 2x : x = 0; \quad y = -\frac{1}{2}x + 5 : x = 10$$

Set $y = 0$ in each line. $y = 2x$ passes through the origin; $y = -\frac{1}{2}x + 5$ crosses the x -axis at $x = 10$. These give the base of the triangle on the x -axis.

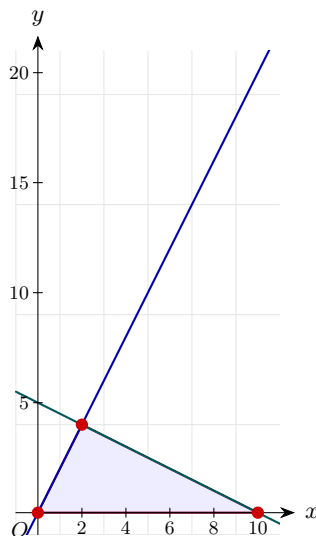
Step 3: Area

$$\text{Area} = \frac{1}{2} \times 10 \times 4 = 20$$

Base along x -axis = 10, height = 4.

Final answer: 20 square units

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 65 (cg3-065)

Difficulty: Foundation / Marks: 5

The perpendicular from $A(0, 4)$ to $y = 2x - 3$ meets the line at B . Find the coordinates of B and the length AB .

Worked solution.

Step 1: Perpendicular from A

$$y - 4 = -\frac{1}{2}(x - 0) \implies y = -\frac{1}{2}x + 4$$

Negative reciprocal of 2 is $-\frac{1}{2}$. Point-slope through $A(0, 4)$; since $x_1 = 0$ the y -intercept is 4 directly.

Step 2: Solve with $y = 2x - 3$

$$2x - 3 = -\frac{1}{2}x + 4 \implies \frac{5}{2}x = 7 \implies x = \frac{14}{5}, y = \frac{13}{5}$$

Equate the two lines. Collect x -terms: $2x + \frac{1}{2}x = \frac{5}{2}x$; the constants give 7. Multiply by $\frac{2}{5}$ to find x , then back-substitute.

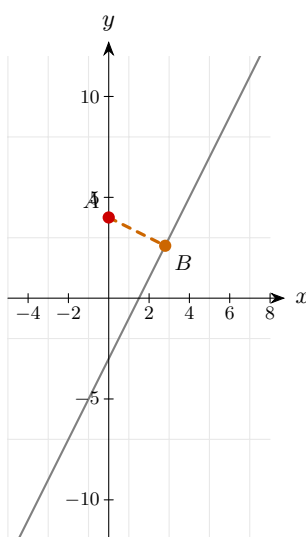
Step 3: Distance

$$AB = \sqrt{\frac{196}{25} + \frac{49}{25}} = \sqrt{\frac{245}{25}} = \frac{7\sqrt{5}}{5}$$

Apply the distance formula: $(\frac{14}{5})^2 = \frac{196}{25}$ and $(\frac{13}{5} - 4)^2 = (-\frac{7}{5})^2 = \frac{49}{25}$. Then $\sqrt{245} = \sqrt{49 \times 5} = 7\sqrt{5}$.

Final answer: $B = \left(\frac{14}{5}, \frac{13}{5}\right); AB = \frac{7\sqrt{5}}{5}$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 66 (cg3-066)

Difficulty: Foundation / Marks: 3

The line L_1 has equation $5x - y = 10$. The line L_2 is parallel to L_1 and passes through $(-1, 3)$. Find where L_2 crosses the y -axis.

Worked solution.

Step 1: Gradient

$$m = 5$$

From $5x - y = 10$: $y = 5x - 10$, so the gradient is 5. Parallel lines share gradients.

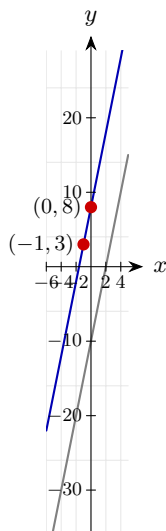
Step 2: L_2 equation

$$y - 3 = 5(x + 1) \implies y = 5x + 8$$

Point-slope through $(-1, 3)$. Expand: $5(x + 1) = 5x + 5$, then add 3. y-intercept is 8.

Final answer: $(0, 8)$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 67 (cg3-067)

Difficulty: Foundation / Marks: 5

Two lines are $L_1 : y = ax + 3$ and $L_2 : y = (2a + 1)x - 5$. Find a if they are: (a) parallel; (b) perpendicular.

Worked solution.

Step 1: (a) Parallel: equal gradients

$$a = 2a + 1 \implies a = -1$$

Setting the gradients equal: subtract a from both sides to get $0 = a + 1$.

Step 2: (b) Perpendicular: product = -1

$$a(2a + 1) = -1 \implies 2a^2 + a + 1 = 0$$

Set the product of gradients equal to -1 . Expand and move all terms to one side to form a quadratic.

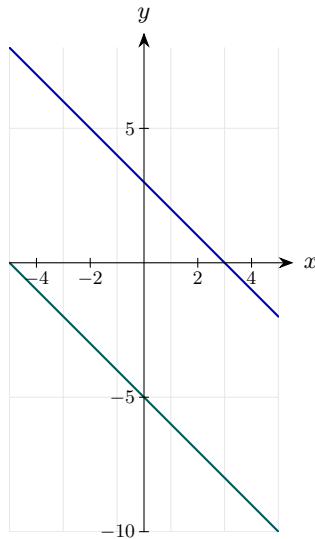
Step 3: Discriminant

$$\Delta = 1 - 8 = -7 < 0$$

No real solutions, so they cannot be perpendicular for any real a .

Final answer: (a) $a = -1$; (b) No real value of a .

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 68 (cg3-068)

Difficulty: Foundation / Marks: 4

Find the equation of the line through the intersection of $y = x + 2$ and $y = 3x - 4$ that is perpendicular to $y = x + 2$.

Worked solution.

Step 1: Find intersection

$$x + 2 = 3x - 4 \implies x = 3, y = 5$$

Equate the two y -expressions: $2x = 6$, so $x = 3$. Substitute back: $y = 3 + 2 = 5$.

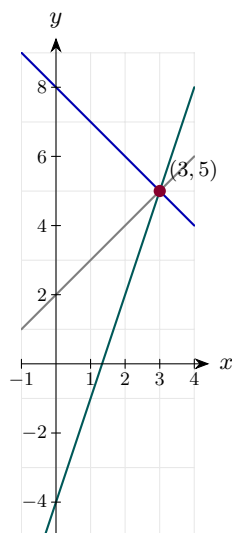
Step 2: Perpendicular to $y = x + 2$

$$m_{\perp} = -1; \quad y - 5 = -(x - 3) \implies y = -x + 8$$

Negative reciprocal of 1 is -1 . Point-slope through $(3, 5)$, then expand and simplify.

Final answer: $y = -x + 8$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 69 (cg3-069)

Difficulty: Foundation / Marks: 4

A line through $(2, 1)$ is perpendicular to the line joining $(-1, 4)$ and $(3, -2)$. Find where it meets the y-axis.

Worked solution.

Step 1: Gradient of line through points

$$m = \frac{-2 - 4}{3 - (-1)} = -\frac{3}{2}$$

Apply the gradient formula. Watch the double negative in the denominator: $3 - (-1) = 4$. Then $\frac{-6}{4} = -\frac{3}{2}$.

Step 2: Perpendicular through $(2, 1)$

$$m_{\perp} = \frac{2}{3}; \quad y - 1 = \frac{2}{3}(x - 2) \implies y = \frac{2}{3}x - \frac{1}{3}$$

Negative reciprocal of $-\frac{3}{2}$ is $\frac{2}{3}$. Expand: $\frac{2}{3}(x - 2) = \frac{2}{3}x - \frac{4}{3}$, then add $1 = \frac{3}{3}$ to get $-\frac{1}{3}$.

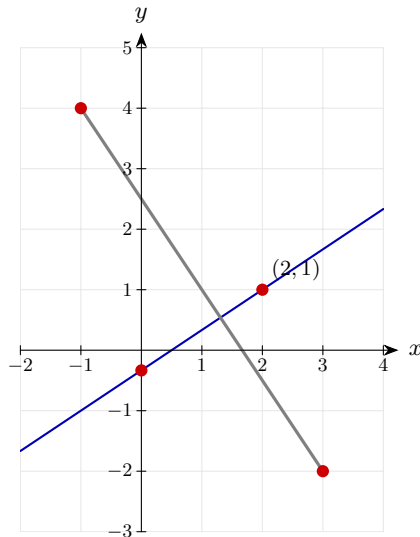
Step 3: y-intercept

$$x = 0 : y = -\frac{1}{3}$$

Substitute $x = 0$ into the line equation.

Final answer: $\left(0, -\frac{1}{3}\right)$

Diagram.



The line(s) and key points shown on the coordinate plane.

Parallel and Perpendicular Lines 70 (cg3-070)

Difficulty: Foundation / Marks: 8

A triangle has vertices $P(1, 1)$, $Q(5, 3)$, $R(3, 7)$. Find: (a) the equation of the altitude from R ; (b) the equation of the altitude from P ; (c) the orthocentre.

Worked solution.

Step 1: (a) Gradient of PQ

$$m_{PQ} = \frac{2}{4} = \frac{1}{2}; \quad m_{\perp} = -2$$

The altitude from R is perpendicular to PQ . Negative reciprocal of $\frac{1}{2}$ is -2 .

Step 2: Altitude from R

$$y - 7 = -2(x - 3) \implies y = -2x + 13$$

Point-slope through $R(3, 7)$. Expand: $-2(x - 3) = -2x + 6$, then add 7.

Step 3: (b) Gradient of QR

$$m_{QR} = \frac{4}{-2} = -2; \quad m_{\perp} = \frac{1}{2}$$

$m_{QR} = \frac{7-3}{3-5} = \frac{4}{-2} = -2$. The altitude from P is perpendicular to QR , giving gradient $\frac{1}{2}$.

Step 4: Altitude from P

$$y - 1 = \frac{1}{2}(x - 1) \implies y = \frac{1}{2}x + \frac{1}{2}$$

Point-slope through $P(1, 1)$. Expand: $\frac{1}{2}(x - 1) = \frac{1}{2}x - \frac{1}{2}$, then add 1.

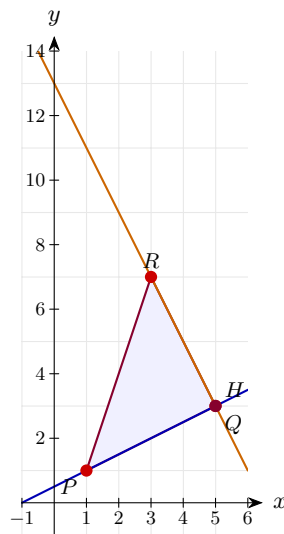
Step 5: (c) Solve altitudes

$$-2x + 13 = \frac{1}{2}x + \frac{1}{2} \implies -\frac{5}{2}x = -\frac{25}{2} \implies x = 5, y = 3$$

The orthocentre is the common intersection of the altitudes. Setting them equal and collecting x -terms: $-2x - \frac{1}{2}x = -\frac{5}{2}x$. Then substitute back to find y .

Final answer: (a) $y = -2x + 13$; (b) $y = \frac{1}{2}x + \frac{1}{2}$; (c) Orthocentre = $(5, 3)$

Diagram.



The line(s) and key points shown on the coordinate plane.
